

When Should We Replace a Predictive Model?

A Finite-Evidence, Refutation-Based Decision Rule

Marcus A. Thomas

Methodological proposal

Abstract

A candidate model that scores better on a benchmark is not yet a model worth deploying in an incumbent’s place. That advantage may depend on which dataset it was measured on, on how common the outcome of interest is there, or on the particular examples that happened to be included—and it can vanish when any of these changes. We ask a stronger question: does a candidate’s measured advantage over an incumbent survive a set of conditions and tests fixed in advance as ones it should have to withstand?

Standard comparisons answer a weaker version of this question. They typically let a strong result in one setting compensate for a weak one in another, and they read the benchmark as a random sample from which to estimate performance on data never collected—though a benchmark need not be such a sample. We develop a decision rule that answers the stronger question from the finite evidence at hand. It aggregates that evidence differently: a candidate is credited only with the advantage that holds up under every declared test on its own, so a gain in one place cannot offset a shortfall in another. Its verdict is what the observed data actually withstood, not an estimate of how the models would compare on data not yet seen. We develop the rule for two widely used ranking measures, average precision and AUROC.

Two illustrations show it in use. On a bank-marketing benchmark, a deliberately strong candidate withstands the tests while a close rival does not. On ProteinGym, a collection of 91 protein mutation-effect assays, released protein language models (ESM-1v, ESM-2) do not withstand enough of these tests to justify replacing an established evolutionary model (EVE).

1 Performance claims and levels of challenge

Prediction systems often rank cases so that a selected set can receive further attention. A radiology service may send its highest-scored images for review; a screening program may investigate its highest-scored records first. The composition of that selected set matters because it is what the decision maker receives.

Separation and selected-set purity part company when prevalence changes. Suppose a threshold selects 80% of positive cases and 5% of negative cases. In a balanced population of 10,000 cases it returns 4,000 positives and 250 negatives, for precision 0.94. In a population with 1% positives, the same threshold returns 80 positives and 495 negatives, for precision 0.14. Its true-positive and false-positive rates have not moved. The supply of negative cases has, and that is enough to change the meaning of the selected set.

Average precision aggregates precision as a ranking is traversed. Ordinary empirical AP therefore evaluates the ranking at the positive fraction in the evaluation data. A performance claim

intended for several deployment populations should therefore name the target prevalences within its scope. No single convenient prevalence establishes a claim throughout a range, and an average over prevalences permits a favorable population to compensate for an unfavorable one.

Call D_m a *full empirical evaluation* when it contains the outcomes and scores needed to evaluate every declared target condition on one set of units. A second opportunity for failure remains inside each full empirical evaluation. A conclusion may depend on the realized combination of represented units. A declared computational procedure can expose that dependence by recombining those units. Call each resulting derived collection a *computational evaluation*. It is distinct from the empirical evaluation from which it is derived, and it is *full* when it contains the outcomes and scores needed to evaluate every declared target condition. It supplies an additional challenge confined to the failure mechanisms represented in the empirical evaluation from which it is derived. The unqualified phrase *full evaluation* refers to either a full empirical evaluation or a full computational evaluation.

A third opportunity for failure arises across distinct full empirical evaluations. An effect retained in one admitted evaluation may fail in another collection of units or context, so a claim intended to survive such tests should declare the regime that admits them.

These three routes of attempted refutation have different scientific identities:

1. a *condition challenge* asks how much of the claimed performance effect survives every declared condition within a full evaluation;
2. a *computational challenge* asks how much of the claimed performance effect survives declared recombinations within one empirical evaluation; and
3. an *empirical challenge* asks how much of the claimed performance effect survives across the evaluations admitted by the evaluation regime.

They share a noncompensation principle: each declared challenge is evaluated as a whole, so a favorable component cannot erase an unfavorable component within that challenge.

The motivating application is model replacement. Candidate A and incumbent B are evaluated on the same units, and larger paired effects favor the named orientation.

At first, this nesting seems to prescribe an inside-out calculation: challenge the conditions, then the computational recombinations, and finally the empirical evaluations. That order would, however, allow derived computational results to enter before the observed empirical evidence has established that there is anything to preserve. Computational recombination can deepen the challenge applied to an empirical evaluation, but it cannot repair an observed empirical failure. The construction therefore reverses the last two steps. It first asks what observed effect survives across the empirical evaluations. Only a claim that survives that empirical challenge is then exposed to the additional within-evaluation computational challenge. This difference between structural nesting and evidential order motivates the observed anchor developed below.

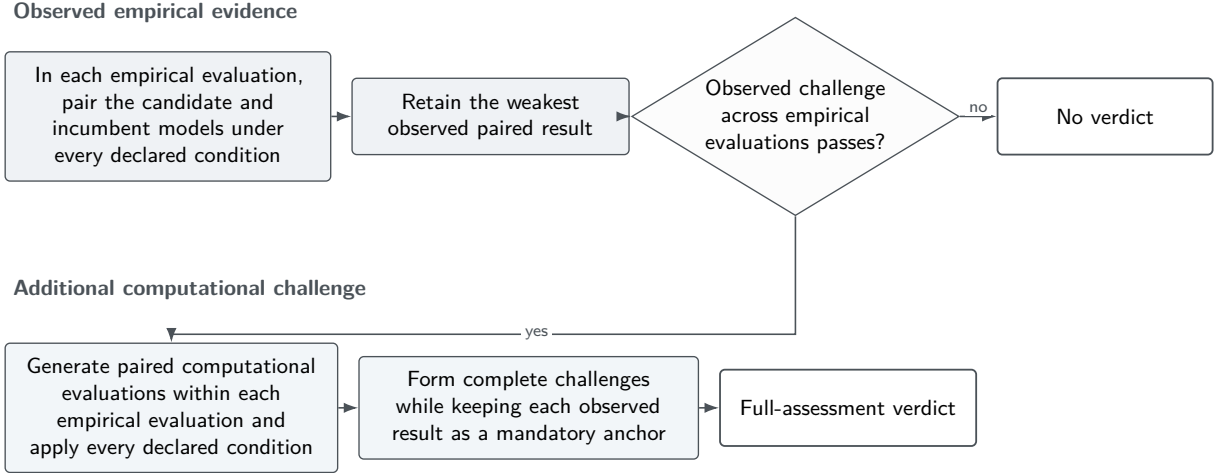


Figure 1: Evidential order of the observed-anchored assessment. Each observed or computational evaluation first faces every declared condition. The observed empirical challenge is a necessary gate. Only a claim that passes that gate proceeds to the additional within-evaluation computational challenge, with each observed result retained as a mandatory anchor.

In this construction, reproducibility is supported by specifying in advance both what the effect is intended to survive and how demanding each challenge will be. Five choices do this. The condition set $\mathcal{C}$ names the conditions examined within every full evaluation. The empirical regime $\mathcal{R}$ determines which observed evaluations are admitted, their evaluation unit and known dependence or multiplicity, whether their identities are retained rather than pooled, and the rule by which they are selected. When a computational challenge is included, $\mathcal{B}$ specifies how represented units are recombined within each empirical evaluation. Finally, K_E and K_C specify the orders of the empirical and computational challenges. We collect these choices in the assessment specification

$$\mathcal{A} = (\mathcal{C}, \mathcal{R}, \mathcal{B}, K_E, K_C).$$

This specification declares the attempted refutations represented by the assessment; it is not a stochastic model for how the observed data or unseen evaluations were generated. Its elements belong to the claim and should be fixed before the results are examined. If no scientifically defensible computational procedure belongs to the assessment, the computational level is absent; Section 3.7 gives the corresponding formal boundary.

The notation used below distinguishes the amount of empirical material or computational approximation available from the severity of the challenge applied to it. The counts N_m , M , and J_m describe what is available; the orders K_E and K_C describe how many effects are jointly faced:

Symbol	Meaning	What increasing it changes
N_m	units in empirical evaluation m	evidence represented within that evaluation
M	admitted empirical evaluations	empirical tests and contexts available
J_m	computational draws within evaluation m	accuracy with which the declared within-evaluation challenge is approximated
K_E	empirical challenge order	number of distinct empirical evaluations jointly faced
K_C	computational challenge order	number of computational effects jointly faced within each selected evaluation

Thus, increasing J_m refines the computational approximation, whereas increasing K_C changes the challenge itself. Likewise, increasing M adds empirical evidence, whereas increasing K_E changes how the available empirical evidence is challenged. No choice of J_m turns one empirical evaluation into several.

Pairing removes the possibility that the two models meet their adverse conditions in different populations or evaluations. It also exposes a further problem: finite-sample AP can reward whichever model divides tied scores more finely, even when that refinement is unrelated to the outcomes.

2 Constructing a retained paired AP effect

The hierarchy will act on effects that have already faced every declared target condition.

2.1 Prior-standardized average precision

Let D contain outcomes and the two fixed score vectors on a common set of N evaluation units. For a generic score S , write the observed class-conditional selection rates at threshold τ as

$$r(\tau) = \mathbb{P}_{\text{obs}}(S \geq \tau \mid Y = 1), \quad f(\tau) = \mathbb{P}_{\text{obs}}(S \geq \tau \mid Y = 0).$$

Prior standardization asks what precision these observed selection rates would produce at a declared target prevalence

$$\pi = \mathbb{P}_{\text{target}}(Y = 1) \in (0, 1).$$

The target population may therefore contain a different proportion of positive cases than the observed evaluation. What is held fixed is the joint behavior of the two model scores within each outcome class. Formally,

$$\mathbb{P}_{\text{obs}}\{(S_A, S_B) \in G \mid Y = y\} = \mathbb{P}_{\text{target}}\{(S_A, S_B) \in G \mid Y = y\}, \quad y \in \{0, 1\}, \quad (1)$$

for every measurable $G \subseteq \mathbb{R}^2$. Here G represents any collection of possible paired scores. The equality says that among positive cases ($Y = 1$), the pair (S_A, S_B) has the same distribution in the observed and target populations, and likewise among negative cases ($Y = 0$). Because the equality holds for every G , it preserves both models' class-conditional score distributions and their

within-class association. Thus, the relative prevalence of the two outcome classes may change, but the paired score behavior within each class does not.

This score-level form of prior shift is weaker than invariance of $P(X | Y)$ but follows from it for fixed scoring rules [25, 28, 17].

Under Equation (1), Bayes' rule gives target precision at threshold τ :

$$p_\pi(\tau) = \frac{\pi r(\tau)}{\pi r(\tau) + (1 - \pi)f(\tau)}. \quad (2)$$

The evaluation supplies r and f ; the declared target population supplies π . Aggregating this precision across the observed positive-class recall increments gives prior-standardized AP, $\text{AP}_\pi(D)$. Appendix A gives the empirical definition. Related calibration methods evaluate precision-based metrics at a fixed reference class ratio [26]. Here π is instead a declared target-population condition and may vary over $\Pi_\star$.

Prior standardization does not transport changes within outcome class. If severity, measurement, or features alter $P(S_A, S_B | Y)$, changing π cannot represent the new context; the framework instead treats it as another empirical evaluation admitted by $\mathcal{R}$ [3].

2.2 A common opportunity scale

Population random ranking has $\text{AP}_\pi = \pi$. Define

$$\text{CNAP}_\pi(D) = \frac{\text{AP}_\pi(D) - \pi}{1 - \pi}. \quad (3)$$

CNAP is zero for population random ranking, one for perfect ranking, and negative below chance. For two models,

$$\text{CNAP}_{A,\pi} - \text{CNAP}_{B,\pi} = \frac{\text{AP}_{A,\pi} - \text{AP}_{B,\pi}}{1 - \pi}. \quad (4)$$

A CNAP margin of a given size therefore requires a raw AP gain equal to that size multiplied by $1 - \pi$ at prevalence π . The scale is appropriate when a replacement margin denotes a fraction of the random-to-perfect opportunity rather than the same raw AP increment in every target population [12].

This linear random-to-perfect normalization differs from precision-recall-gain analysis, which uses harmonic gain coordinates and a different integration measure [15]. CNAP retains the declared target prevalence because selected-set purity in that population is the quantity being challenged.

Common endpoints do not make the profile prevalence-invariant. At a threshold with rates r_h and f_h , the normalized contribution is

$$\frac{p_{\pi,h} - \pi}{1 - \pi} = \frac{\pi(r_h - f_h)}{\pi r_h + (1 - \pi)f_h}. \quad (5)$$

Rare-positive populations charge false positives more heavily. Two models may therefore exchange order as π changes.

Let $\Pi_\star \subset (0, 1)$ be a nonempty compact target-prevalence set selected from scientific knowledge or deployment plans before the paired profile is examined. In the AP application, the condition index is $c = \pi$ and $\mathcal{C} = \Pi_\star$.

2.3 Pair before challenging

A replacement decision concerns an advantage under the same conditions. If the two models are first summarized separately, the adverse prevalence may differ between them and later subtraction will not recover the paired advantage.

For one evaluation, let

$$a = \inf_{\pi \in \Pi_\star} \text{CNAP}_{A,\pi}, \quad b = \inf_{\pi \in \Pi_\star} \text{CNAP}_{B,\pi}, \quad t = \inf_{\pi \in \Pi_\star} \{\text{CNAP}_{A,\pi} - \text{CNAP}_{B,\pi}\}.$$

Then $a - b \geq t$. The separately challenged marginal reports can overstate the advantage because they need not refer to the same prevalence. The same noncomposition persists when effects are combined across evaluations; Appendix C gives the result.

The construction therefore pairs the models at each prevalence before taking the condition infimum.

2.4 Tie-averaged average precision

Ordinary finite-sample AP also depends on how finely a score vector divides ties. Stepwise AP is upward displaced under random ordering because it evaluates precision when positive cases enter the ranking [6]. A fully tied outcome-blind score vector returns the random-ranking value, while an outcome-blind vector of distinct scores creates more opportunities for favorable positive entries. Pairing alone does not remove that advantage.

Let $\mathcal{O}(D)$ contain the strict rankings consistent with a score vector, obtained by resolving each tied block in every possible order while preserving the order between blocks.

Definition 1 (Tie-averaged average precision).

$$\widetilde{\text{AP}}_\pi(D) = \frac{1}{|\mathcal{O}(D)|} \sum_{o \in \mathcal{O}(D)} \text{AP}_\pi(D_o), \quad (6)$$

where D_o is evaluation D under strict ranking o . Define $\widetilde{\text{CNAP}}_\pi$ through Equation (3).

The convention represents outcome-blind random resolution within a score block. It is appropriate when an operational cutoff may split the block and the model supplies no further ordering information. Whole-block selection or a prespecified informative secondary rule would define a different operational regime.

Tie averaging remains responsive to informative refinement. A finer score gains when it orders outcomes correctly within a coarse block and loses when it orders them incorrectly. Only arbitrary refinement receives no expected credit.

Proposition 1 (Neutrality to arbitrary refinement). *Suppose model A preserves the order of model B 's score blocks and, conditionally on S_B and Y , its strict ranking is uniform over the rankings consistent with S_B . Then, for every π ,*

$$\mathbb{E}[\widetilde{\text{AP}}_{A,\pi} \mid S_B, Y] = \widetilde{\text{AP}}_{B,\pi}.$$

A proof is given in Appendix C.2.

The proposition conditions on the realized outcomes, so the coarse model may itself be informative. Appendix D evaluates Equation (6) without enumerating tie resolutions.

2.5 Observed and computational retained effects

The paired effect profile in full evaluation D is

$$\Delta_\pi^{A:B}(D) = \widetilde{\text{CNAP}}_{A,\pi}(D) - \widetilde{\text{CNAP}}_{B,\pi}(D). \quad (7)$$

Within empirical evaluation D_m , the observed retained effect is

$$z_m^{A:B,\text{obs}} = \inf_{\pi \in \Pi_\star} \Delta_\pi^{A:B}(D_m). \quad (8)$$

This is the largest paired advantage retained at every declared target prevalence in the empirical evaluation. Its limiting prevalence may differ between evaluations.

Let $D_m^{(j)}$ be computational replication j generated from D_m under $\mathcal{B}$. Its retained effect is

$$z_{mj}^{A:B,\text{comp}} = \inf_{\pi \in \Pi_\star} \Delta_\pi^{A:B}(D_m^{(j)}). \quad (9)$$

To preserve pairing, the construction uses the same resampled units for both models and throughout the prevalence profile. Resampling separately by model destroys pairing; resampling separately by prevalence splices together conditions from different computational evaluations.

The operations defining one entry are:

pair the models $\longrightarrow$ apply outcome-blind tie handling
 $\longrightarrow$ challenge every target prevalence $\longrightarrow$ retain the weakest paired effect.

This pipeline is written for average precision, but the hierarchy that follows uses only the retained effects it produces, not the metric that produced them. Appendix F instantiates the same construction for AUROC. Because AUROC is unchanged by a shift in the class prior, target prevalence supplies no challenge there; the condition challenge is instead a capped redistribution of the represented cases within each outcome class, governed by a concentration factor $\Gamma \geq 1$ that bounds how far the case mix may be moved. The AUROC analogue of the replacement verdict is a *breakdown factor*: the smallest Γ at which the declared requirement first fails. Average precision and AUROC thus enter one hierarchy through different condition challenges.

3 Support across empirical and computational challenges

The retained effects form a hierarchical array. Its rows are admitted empirical evaluations. Each row has an observed retained effect and a list of computational effects generated from that evaluation:

	observed retained effect	$j = 1$	$j = 2$	$\cdots$	$j = J_m$
D_1	z_1^{obs}	z_{11}^{comp}	z_{12}^{comp}	$\cdots$	$z_{1J_1}^{\text{comp}}$
D_2	z_2^{obs}	z_{21}^{comp}	z_{22}^{comp}	$\cdots$	$z_{2J_2}^{\text{comp}}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
D_M	z_M^{obs}	z_{M1}^{comp}	z_{M2}^{comp}	$\cdots$	$z_{MJ_M}^{\text{comp}}$

(10)

Superscripts identifying $A : B$ are suppressed when the orientation is clear. Every entry has already faced the complete condition challenge.

If a row’s computational effects were allowed to replace its observed retained effect, favorable re-combinations could erase a failure that the empirical evaluation actually exposed. Computation would then be called an additional challenge even though it could make the assessment less adverse. To prevent that reversal, the observed retained effect remains mandatory in every challenge constructed from its row. In that role it is the row’s *observed anchor*.

The remaining problem is to combine anchors and computational effects without allowing computational multiplicity to masquerade as empirical replication.

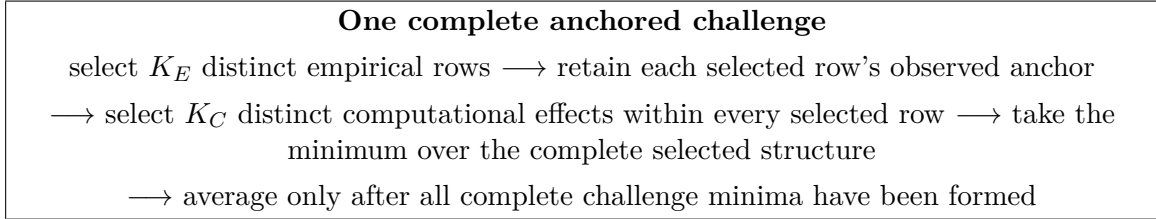


Figure 2: The order of the hierarchical construction. Conditions have already been challenged inside every displayed effect. Empirical rows and computational cells retain their identities throughout the remaining steps.

3.1 The finite-list support operator

Definition 2 (Finite-list survival support). *For a finite list $\mathbf{t} = (t_1, \dots, t_L)$ and $1 \leq K \leq L$, let*

$$\mathcal{I}_{L,K} = \{I \subseteq \{1, \dots, L\} : |I| = K\}.$$

Define

$$S_K(\mathbf{t}) = \binom{L}{K}^{-1} \sum_{I \in \mathcal{I}_{L,K}} \min_{\ell \in I} t_\ell. \quad (11)$$

The entries may be effects from admitted empirical evaluations, empirical replications, or computational replications. Their scientific source and evidential identity should be stated whenever the operator is used.

The operator forms each complete distinct order- K subset, records its weakest effect, and only then averages. At $K = 1$ it is the list mean. At $K = L$ it is the observed minimum. Increasing K moves influence toward the lower entries without asserting a probability law for an unseen list. Every average in this construction is of one kind: a declared counting measure over a realized or constructed-finite collection—here the $\binom{L}{K}$ subsets of the observed list—rather than an assumed law over unrealized data. The distinction lies in the domain of the measure, not in the absence of one, and later sections refer back to it.

For observed anchors $\mathbf{z}^{\text{obs}} = (z_1^{\text{obs}}, \dots, z_M^{\text{obs}})$,

$$\theta_{A:B;K_E}^{\text{obs}} = S_{K_E}(\mathbf{z}^{A:B,\text{obs}}) \quad (12)$$

is direct observed support across the admitted empirical evaluations. It will serve as the empirical gate for the full hierarchy.

3.2 One complete nested challenge

For each selected empirical evaluation m , let $A_m \subseteq \{1, \dots, J_m\}$ contain the K_C distinct computational indices selected from that row, so $|A_m| = K_C$. The weakest effect contributed by that row is

$$\min \left\{ z_m^{\text{obs}}, \min_{j \in A_m} z_{mj}^{\text{comp}} \right\}.$$

Every such rowwise challenge includes the observed anchor. The computational entries add opportunities to expose fragility, but they cannot remove the empirical effect actually obtained.

Now choose K_E distinct empirical evaluations, indexed by I . One complete nested challenge returns

$$\min_{m \in I} \left\{ z_m^{\text{obs}}, \min_{j \in A_m} z_{mj}^{\text{comp}} \right\}.$$

Thus a challenge selects rows, selects computational effects within every selected row, retains the observed anchors, and credits the claim only with the weakest effect anywhere in that selected structure.

A small array makes the order visible:

	z_m^{obs}	z_{m1}^{comp}	z_{m2}^{comp}	z_{m3}^{comp}
D_1	0.30	0.50	0.10	0.40
D_2	0.20	0.25	0.35	-0.05

Take $K_E = 2$ and $K_C = 1$. Retaining the observed anchor from each row and selecting the first computational effect from each row gives

$$\min\{0.30, 0.50, 0.20, 0.25\} = 0.20.$$

Selecting the second effect in D_1 and the third in D_2 gives

$$\min\{0.30, 0.10, 0.20, -0.05\} = -0.05.$$

There are nine complete computational selections. Averaging their anchored minima gives $0.85/9 = 0.094\bar{4}$. By contrast, direct observed order-2 support uses only the two observed anchors:

$$\min\{0.30, 0.20\} = 0.20.$$

This happens to equal the minimum of the first displayed nested selection because the 0.20 observed anchor is the weakest entry in that selection. The computational layer has deepened the challenge without creating more than the two empirical tests represented by the rows.

3.3 Observed-anchored nested support

Definition 3 (Observed-anchored nested support). *Allow empirical evaluation m to contain J_m computational effects. For $1 \leq K_E \leq M$ and $1 \leq K_C \leq \min_{1 \leq m \leq M} J_m$, define*

$$\begin{aligned} S_{K_E; K_C}^{\text{anc}}(\mathbf{z}^{\text{obs}}; \mathbf{Z}^{\text{comp}}) &= \binom{M}{K_E}^{-1} \sum_{I \in \mathcal{I}_{M, K_E}} \left\{ \prod_{m \in I} \binom{J_m}{K_C} \right\}^{-1} \\ &\quad \times \sum_{\substack{(A_m)_{m \in I} \\ A_m \in \mathcal{I}_{J_m, K_C}}} \min_{m \in I} \left\{ z_m^{\text{obs}}, \min_{j \in A_m} z_{mj}^{\text{comp}} \right\}. \end{aligned} \tag{13}$$

The semicolon separates the empirical order from the computational order. For a fixed empirical subset I , row m offers $\binom{J_m}{K_C}$ distinct order- K_C subsets. A complete computational selection chooses one such subset in every row, so I admits

$$\prod_{m \in I} \binom{J_m}{K_C}$$

complete computational selections. For each one, the minimum is taken over the observed anchors and all selected computational effects. Definition 3 averages these complete-selection minima—not separate row summaries—conditional on I and then averages uniformly over the empirical subsets. The conditional normalization prevents an empirical subset from receiving greater weight merely because its rows contain more computational draws.

When every row contains the same number J of computational effects, $J_m = J$, the conditional number of complete selections becomes

$$\prod_{m \in I} \binom{J}{K_C} = \binom{J}{K_C}^{K_E},$$

and Definition 3 reduces to

$$\begin{aligned} S_{K_E; K_C}^{\text{anc}}(\mathbf{z}^{\text{obs}}; \mathbf{Z}^{\text{comp}}) \\ = \frac{1}{\binom{M}{K_E} \binom{J}{K_C}^{K_E}} \sum_{I \in \mathcal{I}_{M, K_E}} \sum_{\substack{(A_m)_{m \in I} \\ A_m \in \mathcal{I}_{J, K_C}}} \min_{m \in I} \left\{ z_m^{\text{obs}}, \min_{j \in A_m} z_{mj}^{\text{comp}} \right\}. \end{aligned}$$

For the oriented paired AP comparison, write

$$\Theta_{A:B; K_E, K_C}^{\text{anc}} = S_{K_E; K_C}^{\text{anc}}(\mathbf{z}^{A:B, \text{obs}}; \mathbf{Z}^{A:B, \text{comp}}). \quad (14)$$

A positive value means that the average weakest paired advantage over complete nested challenges is positive.

The outer factor $\binom{M}{K_E}^{-1}$ in Definition 3 assigns equal weight to every distinct order- K_E empirical subset. Consequently, every admitted empirical evaluation has the same marginal inclusion weight K_E/M , regardless of its number of units, computational draws, or the multiplicity of its broader context in the collection. This uniform empirical weighting is a declared policy choice within $\mathcal{R}$, not a consequence of the finite-evidence interpretation. It prevents larger evaluations from dominating by unit count, but it does not correct contextual multiplicity. A nonuniform empirical weighting would define an extension of the present operator and is not used here.

3.4 Why the hierarchy cannot be flattened or summarized by rows

Flattening Equation (10) into $\sum_m J_m$ computational entries would allow additional Monte Carlo draws from one evaluation to count as additional empirical tests. It would also let the result change merely because one row was assigned a larger approximation count. The empirical index therefore remains intact in the construction.

Nor can the full operator be recovered by first averaging each row:

$$S_{K_E; K_C}^{\text{anc}} \neq S_{K_E} \left(S_{K_C}^{\text{anc}}(z_1^{\text{obs}}; \mathbf{z}_1), \dots, S_{K_C}^{\text{anc}}(z_M^{\text{obs}}; \mathbf{z}_M) \right), \quad (15)$$

where

$$S_{K_C}^{\text{anc}}(u; \mathbf{t}) = \binom{J}{K_C}^{-1} \sum_{A \in \mathcal{I}_{J, K_C}} \min\{u, \min_{j \in A} t_j\}.$$

The right side averages within rows before applying cross-row minima. The full operator instead takes the minimum of every complete nested challenge before averaging.

In the numerical array above, the row summaries are $0.70/3 = 0.233\bar{3}$ and $0.35/3 = 0.116\bar{6}$. Applying order-2 support to those two scalars gives $0.116\bar{6}$, not the correct nested value $0.094\bar{4}$. Averaging and cross-row minimization do not commute.

The same principle governed the paired effect: pair models, challenge conditions, retain anchors, and construct the complete nested challenge before any averaging occurs.

3.5 Literal survival of a declared floor

Supported magnitude retains the numerical minimum from every complete nested challenge. A decision may also require that minimum to clear a declared floor d .

At level s , let

$$R_{m,s} = \sum_{j=1}^{J_m} \mathbf{1}\{z_{mj}^{\text{comp}} > s\}.$$

The fraction of computational order- K_C subsets in row m whose entries all clear s is $\binom{R_{m,s}}{K_C} / \binom{J_m}{K_C}$. A row clears the level only if its observed anchor also clears it. Define the row survival fraction

$$q_{m,K_C,s}^{\text{anc}} = \mathbf{1}\{z_m^{\text{obs}} > s\} \frac{\binom{R_{m,s}}{K_C}}{\binom{J_m}{K_C}}. \quad (16)$$

If the observed anchor fails, no favorable computational subset can rescue the row.

When computational selections are independent across empirical rows, literal survival of the complete nested challenge is

$$\Psi_{K_E; K_C, s}^{\text{anc}} = \binom{M}{K_E}^{-1} \sum_{I \in \mathcal{I}_{M, K_E}} \prod_{m \in I} q_{m,K_C,s}^{\text{anc}}. \quad (17)$$

At the decision floor d , this is the fraction of complete nested challenges in which every selected observed anchor and every selected computational effect exceeds d .

In the running example with $d = 0$, the first row has survival one and the second has survival $2/3$. Because both rows are selected at $K_E = 2$, nested literal survival is $2/3$. The negative computational effect is not averaged against the favorable ones inside the challenges that contain it.

Let e_k denote the elementary symmetric polynomial of degree k . Equation (17) is equivalently

$$\Psi_{K_E; K_C, s}^{\text{anc}} = \frac{e_{K_E}(q_{1,K_C,s}^{\text{anc}}, \dots, q_{M,K_C,s}^{\text{anc}})}{\binom{M}{K_E}}. \quad (18)$$

This form will also make exact finite-array computation possible without enumerating every nested selection.

3.6 Magnitude and survival describe the same challenges

Because the effect array is finite, choose real numbers $a \leq b$ such that every observed and computational effect lies in $[a, b]$. Every complete challenge value t then also lies in $[a, b]$, and

$$t = a + \int_a^b \mathbf{1}\{t > s\} ds.$$

Average this identity over the same nested selections used in Definition 3.

Proposition 2 (Nested magnitude–survival identity).

$$S_{K_E;K_C}^{\text{anc}}(\mathbf{z}^{\text{obs}}; \mathbf{Z}^{\text{comp}}) = a + \int_a^b \Psi_{K_E;K_C,s}^{\text{anc}} ds. \quad (19)$$

A proof is given in Appendix B.2.

Supported magnitude integrates the survival of complete nested challenges across all effect levels. Literal survival evaluates the same curve at the policy floor d . Anchoring one quantity but not the other would sever this identity and make the two decision criteria refer to different challenges.

3.7 Boundary cases and challenge properties

Permit $K_C = 0$ as the mathematical convention

$$A_m = \emptyset, \quad \min_{j \in \emptyset} z_{mj}^{\text{comp}} = +\infty, \quad \binom{J_m}{0} = 1.$$

Then

$$S_{K_E;0}^{\text{anc}} = S_{K_E}(\mathbf{z}^{\text{obs}}), \quad \Psi_{K_E;0,d}^{\text{anc}} = \psi_{K_E,d}(\mathbf{z}^{\text{obs}}), \quad (20)$$

where

$$\psi_{K_E,d}(\mathbf{z}^{\text{obs}}) = \frac{\binom{R_d^{\text{obs}}}{K_E}}{\binom{M}{K_E}}, \quad R_d^{\text{obs}} = \sum_{m=1}^M \mathbf{1}\{z_m^{\text{obs}} > d\}.$$

The phrase *observed-only assessment* is preferable in prose; $K_C = 0$ is not a zero-order empirical replication.

When $M = 1$, $K_E = 1$ is forced and the full construction becomes

$$S_{1;K_C}^{\text{anc}} = \left(\binom{J_1}{K_C} \right)^{-1} \sum_{A \in \mathcal{I}_{J_1,K_C}} \min \left\{ z_1^{\text{obs}}, \min_{j \in A} z_{1j}^{\text{comp}} \right\}. \quad (21)$$

Setting the anchor formally to $+\infty$ recovers the unanchored computational projection. That projection remains a useful secondary diagnostic, but it is not the primary finite-evidence rule developed here.

The principal cases are summarized below.

Setting	Quantity	Role
$M > 1, K_C \geq 1$	$S_{K_E;K_C}^{\text{anc}}$	full empirical–computational assessment
$M = 1, K_E = 1, K_C \geq 1$	$S_{1;K_C}^{\text{anc}}$	anchored one-evaluation assessment
$K_C = 0$	$S_{K_E}(\mathbf{z}^{\text{obs}})$	empirical gate; reduced assessment if computation is omitted
anchor = $+\infty$	unanchored construction	secondary computational projection

Every complete anchored challenge contains its observed anchors. Consequently, the full construction cannot exceed the observed-only analysis.

Proposition 3 (Anchored challenge domination). *For every admissible K_E, K_C and floor d ,*

$$S_{K_E;K_C}^{\text{anc}}(\mathbf{z}^{\text{obs}}; \mathbf{Z}^{\text{comp}}) \leq S_{K_E}(\mathbf{z}^{\text{obs}}), \quad (22)$$

$$\Psi_{K_E;K_C,d}^{\text{anc}} \leq \psi_{K_E,d}(\mathbf{z}^{\text{obs}}). \quad (23)$$

The anchored quantities also do not exceed the otherwise identical unanchored nested projection.

A proof is given in Appendix B.3.

The condition set and both orders control different dimensions of severity.

Proposition 4 (Hierarchical challenge monotonicity). *Subject to the required list lengths, anchored support and literal survival at a fixed floor cannot increase when $\mathcal{C}$ is enlarged, K_C is increased, or K_E is increased.*

A proof is given in Appendix B.4.

The regime $\mathcal{R}$ has no such mechanical ordering. Changing it changes the scientific scope of the empirical tests rather than merely strengthening a fixed challenge.

3.8 Orientation

The reverse profile is $\Delta_{\pi}^{B:A}(D) = -\Delta_{\pi}^{A:B}(D)$, but negation occurs before the condition infima and all nested minima. The reverse anchored value is therefore not generally the negative of the forward value.

Proposition 5 (Nested orientation bound). *When both orientations use the same evaluations, computational replications, and nested selections,*

$$\Theta_{A:B;K_E,K_C}^{\text{anc}} + \Theta_{B:A;K_E,K_C}^{\text{anc}} \leq 0. \quad (24)$$

A proof is given in Appendix B.5.

The construction cannot support positive challenged advantages in both directions. Both may be nonpositive because different conditions, empirical evaluations, or computational effects defeat the two directional claims.

Reference behavior may be examined through a deliberately constructed exchangeable-label randomization. Under full exchangeability and outcome-blind tie handling, the expected condition-specific paired effect is zero and the expected anchored supported value is nonpositive. This is a finite randomization reference conditional on the observed data, used descriptively rather than as a presumed sampling law. Appendix C defines the randomization, gives the finite-list argument, and shows why restricted label symmetries need not have the same anchor.

4 A staged finite-evidence replacement rule

The operator describes what survived the declared challenges. A fully specified replacement policy also states how much retained advantage and how much literal survival are sufficient.

A familiar way to fix such thresholds would minimize an expected loss: assign a loss to each action, average it over the states the world might take, and choose the action of smallest average loss. That procedure needs a probability over unobserved states. The rule below never averages over unobserved states. It instead fixes, in advance, which pattern in the *observed* evidence would count as refuting the claim, and then reports whether the observed evidence exhibits that pattern. The thresholds are the terms of that advance declaration, not the coefficients of a loss integral; this is the conjecture-and-refutation stance of Section 1 made quantitative [23, 24].

Let $\delta \geq 0$ be the smallest supported magnitude that would justify replacement. Let $d \geq 0$ be the floor used to define literal survival, and let $\gamma \in (0, 1)$ be the minimum acceptable survival fraction. These are policy commitments: terms of the refutation criterion, fixed before the array is examined, rather than confidence levels or quantities learned from the effect array.

The thresholds and challenge orders play three distinct roles. The floor d and the magnitude level δ fix the *content* of the conjecture: d is the per-evaluation advantage the claim asserts—at $d = 0$, strictly better than the incumbent—and δ is the aggregate supported magnitude it commits to. A bolder claim raises d or δ , and boldness means more that could go wrong, hence a claim that is easier to refute. The orders K_E and K_C fix the *stringency* of aggregation: they control how a single failure inside a challenge compounds, from a tolerant average at order one to an unforgiving worst case at the maximal order. The fraction γ fixes the required *breadth*: how much of the actual evidence must show clean survival. Breadth and stringency are not independent knobs. As Proposition 6 shows, at the observed gate γ and K_E reduce jointly to a single statement about the number of refuting evaluations the claim tolerates.

A reader meeting these thresholds for the first time may import a probabilistic model that none of them carries. The following contrast names what each parameter actually declares.

Parameter	A reader may import	What it actually declares
d	an error tolerance	the per-evaluation advantage the claim asserts
δ	a required expected effect size	the aggregate supported magnitude the claim commits to
γ	a probability or confidence of success	the fraction of the actual evidence that must show clean survival
K_E, K_C	a sample size or count of replications	how failures compound within a single challenge

4.1 Stage 1: the observed empirical gate

The oriented claim passes the empirical gate only if

$$\theta_{A:B;K_E}^{\text{obs}} > \delta \quad \text{and} \quad \psi_{A:B;K_E,d}^{\text{obs}} > \gamma. \quad (25)$$

The magnitude requirement prevents a negligible but widespread advantage from automatically justifying replacement. The survival requirement prevents large favorable effects from compensating for too many observed failures at d .

Proposition 6 (Tolerated refuting evaluations at the observed gate). *Let $R = \sum_{m=1}^M \mathbf{1}\{z_m^{\text{obs}} > d\}$ be the number of observed anchors clearing the floor. Observed literal survival $\psi_{K_E,d}^{\text{obs}} = \binom{R}{K_E} / \binom{M}{K_E}$ is nondecreasing in R and strictly increasing for $R \geq K_E - 1$. Therefore $\psi_{K_E,d}^{\text{obs}} > \gamma$ holds if and only if*

$$R \geq r^*, \quad r^* = \min \left\{ r : \binom{r}{K_E} > \gamma \binom{M}{K_E} \right\}.$$

Equivalently, the pair (K_E, γ) declares that at most $M - r^$ of the M admitted evaluations may be refuting instances.*

A proof is given in Appendix B.6.

The survival fraction is therefore a disguised count. It states, before the array is examined, how many falsifying evaluations the claim tolerates while still counting as unrefuted—a declaration with no probabilistic content. The ProteinGym illustration of Section 6.2 is exactly this boundary: with $K_E = 2$ and $\gamma = 0.81$ over $M = 91$ assays, $r^* = 82$, so the criterion tolerates at most nine refuting assays. The equivalence is exact at the observed gate. Once a computational challenge enters ($K_C \geq 1$), literal survival becomes the elementary symmetric polynomial in the fractional row survivals $q_{m,K_C,s}^{\text{anc}}$ of Equation (16), and no single tolerated count reproduces it; the proposition is a statement about Stage 1.

Failure of either condition returns no verdict. It does not establish that the incumbent is superior. By anchor domination, computation cannot reverse the failure.

4.2 Stage 2: the computational challenge

If Stage 1 passes and $\mathcal{B}$ belongs to the declared assessment, the primary full rule requires

$$\Theta_{A:B;K_E,K_C}^{\text{anc}} > \delta \quad \text{and} \quad \Psi_{A:B;K_E,K_C,d}^{\text{anc}} > \gamma. \quad (26)$$

Passing the observed gate but failing Equation (26) means that the advantage obtained across the empirical tests did not survive the declared computational recombinations of their represented units. Passing both stages supports replacement under the full anchored assessment.

The sequence is an ordered set of attempted refutations:

$$\begin{aligned} &\text{paired condition challenge} \longrightarrow \text{observed empirical gate} \\ &\longrightarrow \text{within-evaluation computational challenge} \longrightarrow \text{full-assessment verdict.} \end{aligned}$$

The observed gate is reported even when Stage 2 is primary because it shows what the empirical evaluations establish directly and identifies where the claim was defeated.

4.3 Reduced assessment and reverse orientation

If no scientifically defensible $\mathcal{B}$ is available, or if omission of the computational level was prespecified for a practical reason, the analysis may stop at Equation (25). A passing verdict then applies to the reduced observed-only assessment. It should not be described as passage of the full anchored rule.

Computational infeasibility identified in advance can justify declaring a reduced observed-only assessment. If Stage 2 belongs to the declared assessment and causes the full rule to return no verdict, the analysis should not subsequently omit Stage 2 and report the passing Stage 1 result as support under a reduced assessment. Nor should a smaller positive J_m be described as a substantively weaker challenge; it is a noisier approximation to the same declared target.

The reverse comparison is constructed from $\Delta_\pi^{B:A} = -\Delta_\pi^{A:B}$ before all condition infima, anchors, and nested minima. It faces the same staged rule, with orientation-specific thresholds if the consequences are asymmetric. If neither direction passes, the result is no verdict.

The rule applies to the finite evidence and attempted refutations identified in $\mathcal{A}$. It does not estimate a probability that an unseen evaluation will pass, and it does not convert a computational law into a law of nature. Using the verdict for deployment requires an external premise that the declared conditions, regime, resampling procedure, and transport commitments are adequate for the contemplated use. The calculation exposes those commitments; it does not certify them.

5 Evidence and computational implementation

The hierarchy preserves a difference that no algebra should erase. Empirical evaluations supply evidence. Computational replications act on that evidence under a declared procedure. The same numerical effect may appear at either level, but its index records a different attempted refutation.

5.1 Defining the empirical evidence

The regime $\mathcal{R}$ specifies which complete observed evaluations count as tests of the claim. It identifies the object under test, eligible populations, measurement standards, required outcomes and scores, empirical evaluation unit, known dependence or multiplicity, and selection or pooling rule. A full observed evaluation satisfying these requirements is an *admitted empirical evaluation*. Admission places it in the declared finite evidence; it does not imply independence or sampling from a common population.

An admitted evaluation is an *empirical replication* only when it was separately conducted, its units are disjoint from those of the other replications, and its model construction and data collection followed a protocol fixed without using their results. Resampling draws, administrative partitions, cross-validation folds, and repeated analyses of the same units remain analyses within one empirical evaluation. Evaluations that prompt model or protocol changes instead form an adaptive sequence.

Evaluation identities should be retained when contextual variation is part of the scoped claim. Pooling may be appropriate when dataset membership is merely administrative and the units address one aggregate population. Thus, $\mathcal{R}$ determines which empirical tests are represented and how they are counted; it does not supply a probability law over unseen evaluations.

5.2 Defining the computational challenge

The procedure $\mathcal{B}$ specifies how computational evaluations are formed by recombining units represented within each empirical evaluation. It declares the resampling unit, draw size, stratification or block structure, and the treatment of draws lacking required outcome classes. In every computational evaluation, the outcome and both model scores remain attached to each selected unit or block, and the same realized selection is used for both models and throughout the condition profile. Each retained computational effect therefore comes from one complete paired evaluation.

The procedure determines which dependence on the represented units is challenged. Observation-level resampling may sample separately within outcome classes and preserve declared class counts. A clustered or matched design instead resamples the declared design blocks; the resulting class counts and number of component observations may vary. These procedures define different computational challenges, and the choice between them should follow the scientifically actionable unit.

The computational law assesses whether the observed advantage depends on the particular represented units without changing the empirical scope. If empirical evaluations have nonoverlapping units and are resampled independently, computational selections across rows give the product in Equation (17). Shared units or coordinated perturbations require a joint computational law that preserves the relevant dependence. The direct challenge definition remains valid in that case, but the product representation may not.

5.3 The empirical gate is an exact stopping rule

Calculate the observed anchors and their direct order- K_E summaries before generating computational replications for the primary verdict:

$$\theta_{A:B;K_E}^{\text{obs}} = S_{K_E}(\mathbf{z}^{A:B,\text{obs}}), \quad \psi_{A:B;K_E,d}^{\text{obs}} = \psi_{K_E,d}(\mathbf{z}^{A:B,\text{obs}}).$$

If either observed quantity fails its policy threshold, Proposition 3 proves that the full anchored rule also fails. The primary analysis can stop with no verdict. If the empirical gate passes, generate the computational effects within each row and evaluate the full anchored quantities.

5.4 Computational target and Monte Carlo approximation

The finite array approximates a functional under the declared resampling law. If one computational retained effect in row m exceeds s with probability $p_m(s)$ under that law, then the idealized iid order- K_C row survival is

$$q_{m,K_C,s}^{\text{anc},\infty} = \mathbf{1}\{z_m^{\text{obs}} > s\} p_m(s)^{K_C}. \quad (27)$$

The finite distinct-subset fraction

$$\binom{R_{m,s}}{K_C} / \binom{J_m}{K_C}$$

is a U-statistic approximation to that row target. This distinction explains why K_C is substantive while J_m is numerical. Lowering K_C weakens the declared challenge; lowering J_m makes the same target less accurately approximated.

When the condition challenge is defined over a continuous set, its infimum also requires numerical approximation. Missing an interior minimum can raise a computed retained effect, so numerical

accuracy of the effect values should be checked separately from Monte Carlo stability. Appendix E gives the exact finite-array algorithm and the corresponding numerical checks.

6 Empirical illustrations

Two illustrations occupy different positions in the hierarchy. Bank Marketing begins with one temporal empirical evaluation. Its positive-control comparison passes the observed gate and proceeds to the anchored computational challenge; its close model comparison fails at the gate. ProteinGym supplies 91 observed assay evaluations, but all examined orientations fail the observed gate. The second stage is therefore unnecessary for their verdicts.

Bank Marketing uses $\Pi_\star = [0.01, 0.50]$; ProteinGym uses $\Pi_\star = [0.01, 0.90]$. Both use $\delta = d = 0$ and $\gamma = 0.81$. Bank Marketing uses $K_E = 1$ and, when Stage 2 is reached, $K_C = 2$. ProteinGym uses observed order $K_E = 2$. These choices define the illustrated policies and are not proposed as defaults.

6.1 One empirical evaluation: Bank Marketing

The UCI Bank Marketing data contain ordered direct-marketing contacts and whether each customer subscribed to a term deposit [20]. The first 32,950 of 41,188 records were used to fit the models; the final 8,238 formed one temporal empirical evaluation. Three fixed model specifications were fitted. The full logistic regression and random forest used all available pre-contact predictors and formed the close comparison. The demographic logistic regression used only age, job, marital status, and education and provided a deliberately larger positive-control contrast with the full logistic model. Categorical predictors were represented by one-hot indicators. Contact duration was excluded because it is unavailable when the contact decision is made. All reported scores came from the temporal holdout. These models and the split illustrate the method rather than support a claim about banking practice.

Figure 3 holds the full-logistic ranking and its observed class-conditional score behavior fixed while varying target prevalence. AP changes with the target composition. CNAP keeps population random ranking at zero while retaining a nonconstant profile.

The paired model difference is therefore formed before the weakest prevalence is selected. Figure 4 shows that the random-forest ordering changes relative to full logistic regression inside the declared interval, whereas the positive-control difference remains positive throughout.

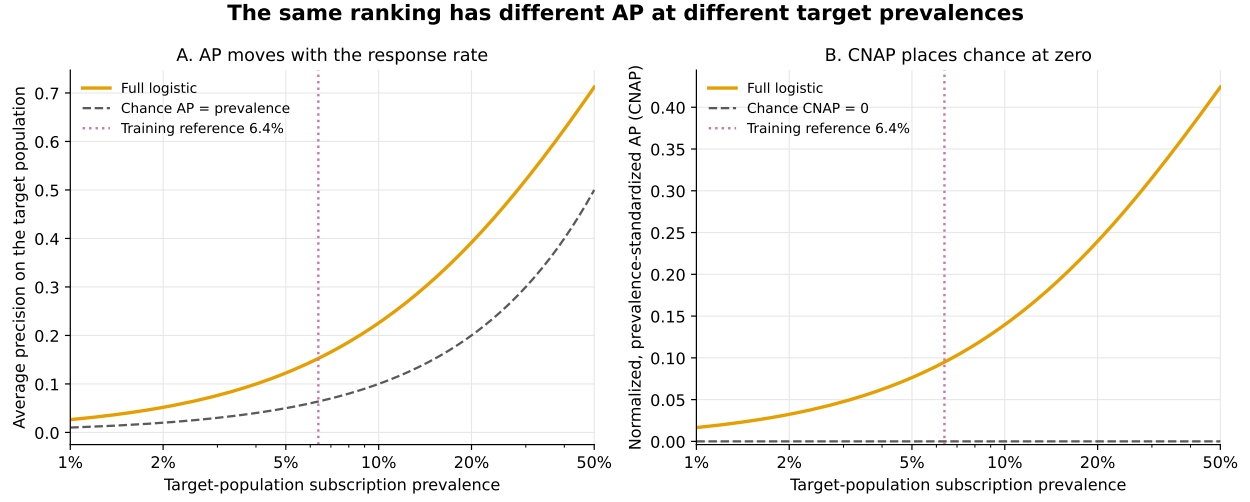


Figure 3: Prior-standardized AP and CNAP for one fitted ranking across the declared target-prevalence interval. The training prevalence is shown only as a reference. Varying prevalence holds the observed class-conditional score behavior fixed.

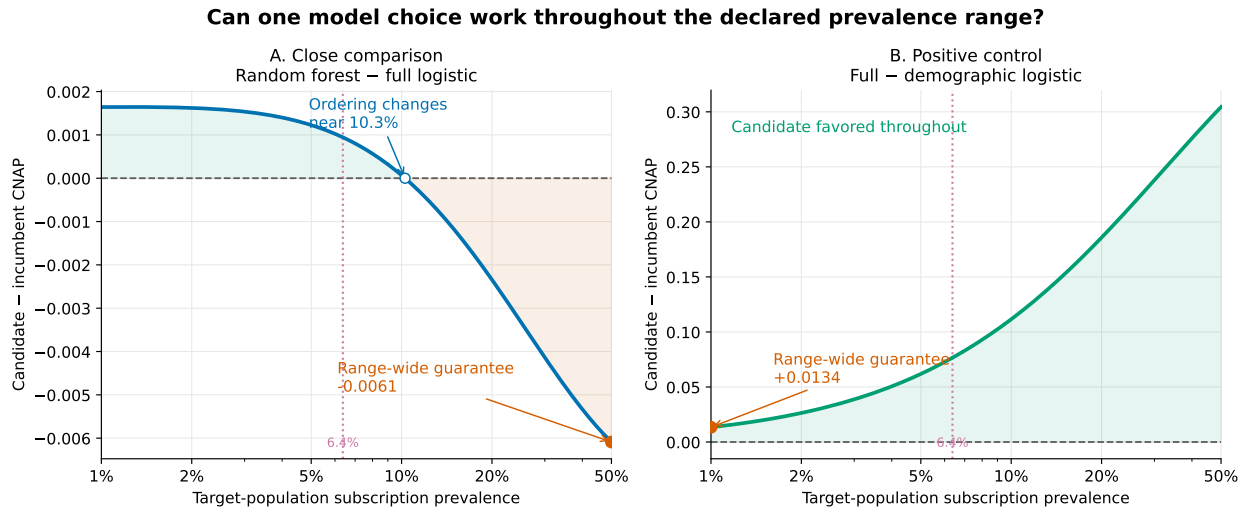


Figure 4: Paired CNAP differences in the temporal empirical evaluation. Panel A compares random forest with full logistic regression; Panel B compares full logistic regression with the demographic-only logistic model. Each orange point is the infimum over the complete declared prevalence interval. The limiting prevalence may differ between comparisons and orientations.

The temporal holdout forces $M = 1$ and $K_E = 1$. Random forest minus full logistic regression had observed retained effect -0.00609 ; the reversed orientation had retained effect -0.00164 . Both fail the observed gate, so anchor domination makes a full anchored verdict impossible. The construction therefore stops at Stage 1: no computational AP collection is generated for this comparison.

The positive control had observed retained effect 0.01341 ; its reverse retained effect was -0.30442 . The forward orientation therefore passed the observed gate for the single evaluation. We then used $J = 600$ paired computational replications stratified by outcome class, preserving the observed evaluation counts of 2,540 subscribers and 5,698 non-subscribers. The same resampled units were used for both models and throughout the prevalence profile.

For the positive control, observed-anchored order-2 support was 0.01262 . All 600 computational retained effects and the observed anchor cleared zero, so anchored literal survival was 1.000 . The comparison passed Stage 2 and supported replacement under the full one-evaluation assessment. The unanchored order-2 projection was 0.01295 ; its slightly larger value is reported only to show the effect of retaining the observed anchor. The reverse orientation returned no verdict.

This example establishes computational survival conditional on one temporal evaluation. It records no survival across several empirical evaluations.

The corresponding observed-anchored AUROC case-mix analysis is reported in Appendix F.3. It uses the same temporal holdout and paired model comparisons but challenges within-class composition rather than target prevalence.

6.2 Several empirical evaluations: ProteinGym

ProteinGym organizes its deep-mutational-scanning (DMS) benchmark into standardized assay-indexed datasets containing experimentally measured effects for variants of a target protein or domain under a particular experimental readout and context, together with released prediction scores [22]. The task here is to rank single substitutions so that experimentally deleterious or non-functional variants appear ahead of functional variants. Such rankings can prioritize substitutions for further interpretation or experimental study when exhaustive measurement is unavailable.

We compared released zero-shot scores from three methodologically distinct protein sequence models: the EVE ensemble evolutionary model, the ESM-1v ensemble protein language model, and the ESM-2 650M protein language model [16, 19, 18]. The comparisons were ESM-1v ensemble against EVE ensemble, ESM-2 650M against ESM-1v ensemble, and ESM-2 650M against EVE ensemble. No model was fit or tuned for this study. The replacement question was whether one released ranking retained a positive advantage throughout the target deleterious-prevalence interval and across a sufficiently large fraction of the admitted datasets.

We treated each retained assay-indexed dataset as one empirical evaluation. Within a dataset, the substitutions supplied the paired model scores and experimental labels used to compare the rankings; they were observations within an evaluation, not separate empirical replications. This operational choice preserves the standardized units supplied by ProteinGym and prevents datasets containing more substitutions from dominating, but it does not claim that an assay row is the uniquely correct or a statistically independent unit of scientific replication. The across-evaluation challenge therefore asked whether the average advantage recurred across these dataset contexts. Greater domain knowledge could motivate another prospectively declared operationalization—for example, grouping assays by protein, publication, experimental system, or shared measurement

context—which could change both the effective collection size and the interpretation of across-evaluation support.

We used ProteinGym version 1.3 substitution data, retained assays with manual biological cutoffs, and retained single amino-acid substitutions. Experimentally nonfunctional or deleterious variants formed the positive class. Because the released scores were oriented toward predicted fitness, they were negated so that larger analyzed values represented stronger predicted deleteriousness. The aligned cohort contained 277,873 substitutions in 91 assays, including 96,987 deleterious and 180,886 functional variants.

The observed deleterious fraction ranged from 0.025 to 0.869 across assays, with median 0.315 and interquartile range [0.244, 0.440]. All 91 assay fractions lay within the declared target interval [0.01, 0.90]. The pooled fraction was $96,987/277,873 = 0.349$, but this value weights assays by their numbers of retained substitutions and is not the prevalence of a typical assay.

Ordinary empirical AP would evaluate each ranking at the deleterious fraction observed in that assay. Prior standardization instead evaluates every assay over the common target interval while retaining that assay’s observed joint class-conditional score behavior. It changes only the target class prior; each assay’s observed within-class score behavior and its protein, measurement process, and experimental context are preserved.

Within each assay, the paired CNAP difference was evaluated throughout the complete interval and its smallest value was retained as the observed anchor. A positive anchor therefore means that the candidate exceeded the incumbent at every declared target prevalence in that assay. Figure 5 shows positive and negative anchors in every comparison, so the relative model ordering was not stable across the admitted assays.

Table 1 reports the Stage 1 challenge across the 91 assay anchors. With $K_E = 2$, each complete empirical challenge contains two distinct assays and survives the floor $d = 0$ only when both anchors are positive.

Table 1: Stage 1 observed order-2 ProteinGym results. All comparisons use the same 91 assays and policy.

Candidate minus incumbent	Support	Assays > 0	Pair survival	Gate
ESM-1v ensemble minus EVE ensemble	−0.1548	49/91	0.287	Fail
ESM-2 650M minus ESM-1v ensemble	−0.1175	39/91	0.181	Fail
ESM-2 650M minus EVE ensemble	−0.1556	53/91	0.337	Fail

An order-2 pair contains two complete assay evaluations, not two protein variants. There are $\binom{91}{2} = 4,095$ such pairs, and a pair survives at $d = 0$ only when both assay anchors are positive. The strict requirement $\psi_{2,0} > 0.81$ requires at least 82 of 91 assays to clear zero because $\binom{82}{2}/\binom{91}{2} = 0.811 > 0.81$; this is the tolerated-count reading of Proposition 6, with $r^* = 82$. Thus more than 81% of all distinct assay pairs survive only if at least 82 of 91 individual assays—just over 90%—retain a positive range-wide advantage. None approached that count.

All forward and reverse comparisons fail the empirical gate. This does not show that an incumbent was superior in every assay or that the compared models were equivalent. It shows that no examined orientation produced a positive range-wide advantage in enough assay evaluations to satisfy the declared replacement policy.

Advantage retained across 1%–90% deleterious prevalence

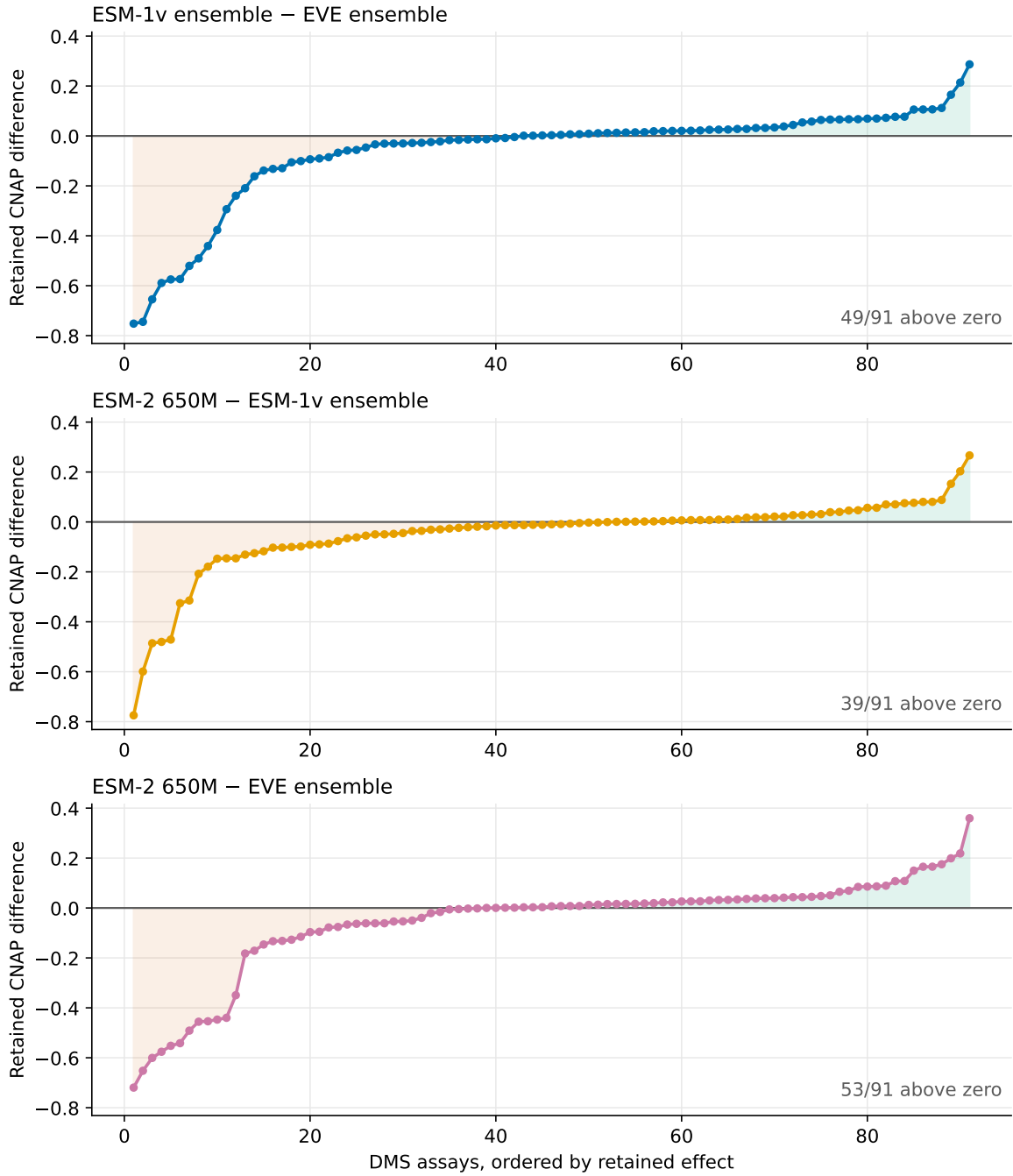


Figure 5: Observed retained assay effects for the three ProteinGym comparisons over the 1%–90% target-prevalence interval. Each point is one complete DMS assay after its paired difference has faced the full interval. Panels order assays separately; the horizontal position does not identify the same assay between panels.

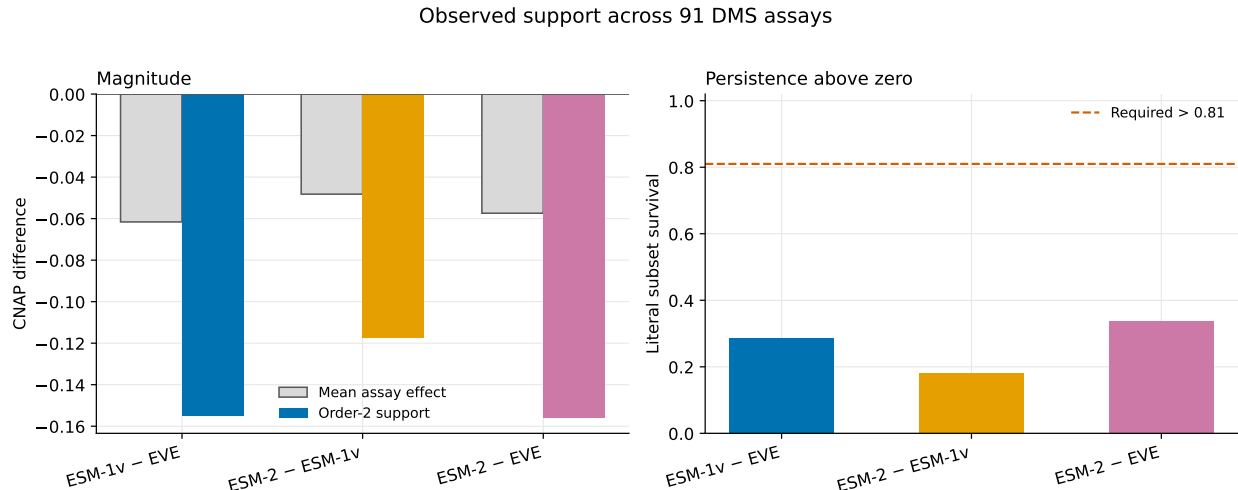


Figure 6: Observed mean assay effect, order-2 support, and literal assay-pair survival for the three ProteinGym comparisons. The left panel shows how the noncompensating order-2 challenge moves magnitude below the assay mean. The bars on the right are exact fractions of the 4,095 assay pairs whose two anchors both exceed zero.

Because the observed assay anchors fail the necessary gate, resampling substitutions within assays could not support replacement under the anchored rule. Such resampling might describe sensitivity to the represented substitutions, but it could neither convert an observed assay failure into a success nor add new proteins, assays, laboratories, or measurement mechanisms. We therefore do not resample merely to fill the interior $M > 1, K_C > 0$ case. Doing so would also require a defensible account of within-assay dependence, protein-position clustering, sparse outcomes, and the scientific identity of the resampling unit.

Appendix E.3 nevertheless uses a deliberately favorable, post hoc ProteinGym specification to illustrate the full $M > 1, K_E = K_C = 2$ calculation: even after restricting the prevalence interval to clear Stage 1, the observed-anchored computational challenge returns no verdict at Stage 2.

The 91 assay-indexed datasets represent 80 UniProt proteins and 60 publications. Twenty assay rows share a protein with another retained assay, and 42 come from publications contributing more than one assay. We retained these multiplicities in the finite collection rather than grouping by protein or publication. Under the declared assay-level regime, target interval, empirical order, and policy thresholds, none of the examined orientations supports model replacement. The result concerns the 91 admitted assay evaluations; it does not establish global model superiority or estimate performance in an unseen assay.

Appendix F.3 also reports the observed-only AUROC boundary for these same 91 assays. Because every declared comparison fails that empirical gate, no within-assay computational AUROC collection is generated.

7 Interpretation, reporting, and scope

7.1 The finite-evidence basis for model replacement

Observed support records what the admitted empirical evaluations retain after the condition challenge. Anchored empirical–computational support records what remains after their observed anchors face the declared computational challenge. Both are indexed by the orientation, condition set, evaluation regime, challenge orders, and—for the full construction—the computational procedure.

This basis is falsification-oriented: replacement is supported when the candidate’s advantage survives the opportunities for failure specified in advance as relevant to the decision [23, 24]. The objective is not to estimate a probability that the result will replicate in a future evaluation. Such a probability requires a defined population of possible evaluations, a rule for sampling from it, and a decision about how different contexts should be weighted. A finite benchmark collection generally supplies none of these: its evaluations may be curated, heterogeneous, dependent, or selected because they represent particular scientific contexts. Without an additional population model, a future-replication probability is therefore undefined rather than merely estimated imprecisely.

More importantly, averaging over a hypothetical distribution of future evaluations would answer a different question. The replacement rule asks whether the candidate’s advantage survived the conditions, empirical evaluations, and computational challenges declared as relevant to the present decision, without allowing favorable settings to compensate for specified failures. The finite empirical list is therefore treated as a collection of realized tests, and literal survival is the exact fraction of its declared challenges that clear the floor, not a confidence level. The resulting verdict supports a ranking-performance decision within that declared scope; calibration, utility, resource constraints, and consequences of deployment may be incorporated separately when the practical decision requires them.

7.2 Relation to population robustness and inference

Some operations have analogues in frameworks that begin with a population probability law. Shared algebra does not supply a shared interpretation.

The retained effect $\inf_{c \in \mathcal{C}} \Delta_c(D)$ resembles an intersection bound. Intersection-bound inference instead concerns infima or suprema of population-indexed functions and their repeated-sampling behavior [11]. Here the infimum extracts the exact uniform lower level of a condition profile computed in one complete finite evaluation.

The ordered finite-list form of $-S_K$ resembles a spectral loss functional [1], and the survival integral resembles a distortion or Choquet representation [31]. In those settings, order statistics commonly estimate a functional of a random quantity. Here the representations are exact identities for a declared finite indexed list or array. No superpopulation interpretation is imported.

The empirical case-mix sets developed in Appendix F resemble ambiguity sets in distributionally robust optimization [27, 10]. Their concentration factor is a deterministic allowance over represented cases, not a confidence radius around an unknown distribution. Population coverage results from data-driven robust optimization require a different inferential target and assumptions [14].

Probability is used here when a target population description or deliberately constructed randomization requires it. The target prevalence describes a counterfactual population composition. The

resampling law defines a computational projection. The exchangeable-label law defines a reference diagnostic. None is silently promoted into a law generating unseen empirical evaluations.

Inferential comparisons of classifiers across repeated train–test samples or multiple benchmark datasets instead ask whether performance differences generalize under sampling assumptions [21, 7, 13]. The empirical subset average here is an exact finite selection over admitted evaluations and supplies no sampling law for unseen datasets.

Two readings deserve explicit separation, because the survival fraction invites them. A p -value integrates a statistic over unrealized samples drawn from an assumed law; the survival fraction is a census of realized challenges under the declared uniform measure over the $\binom{M}{K_E}$ subsets of the actual anchors. Nothing unrealized enters, so the two are different objects despite both lying in $[0, 1]$.¹ The framework accordingly offers no inferential calibration of γ : it does not state how often a claim clearing the gate would clear it again on new evaluations, and supplying such a figure would require the population law the construction declines to assume. The admissible justification of a threshold is the stakes of the replacement decision and the stringency intended, never the replication it predicts.

7.3 On cross-validation and generalization

We use *generalization* in a broad sense: persistence of a conjecture beyond the evidence from which it was constructed, throughout the circumstances included in its claimed scope. The familiar predictive definition is a special case. If the data-generating process is stipulated to produce exchangeable observations from a fixed law P , and the intended scope consists of further draws from that same law, then generalization may be represented by expected predictive loss,

$$R_P(A) = \mathbb{E}_{D_{\text{train}}, Z_{\text{new}} \sim P} [L\{A(D_{\text{train}}), Z_{\text{new}}\}],$$

for learning procedure A . Ordinary cross-validation is most coherent under this conception: it approximates recurrence of predictive performance across training samples and future observations governed by the stipulated law [2, 5]. Structured splitting may replace exchangeable folds when dependence, time, or grouping is acknowledged, but the interpretation still requires the splitting scheme and population law to represent the recurrence of scientific interest.

The finite cross-validation calculation should nevertheless be distinguished from that interpretation. If $\ell_i = L\{y_i, \hat{f}_{-k(i)}(x_i)\}$ is the loss for observation i under a fit that withheld its fold, then

$$\hat{R}_{\text{CV}} = \frac{1}{n} \sum_{i=1}^n \ell_i$$

is an exact average of realized discrepancies. Reading it as an estimate of $R_P(A)$ adds a law over unrealized observations and training samples. That population interpretation is an additional premise, logically separate from the withholding calculation.

That separation matters because, outside simulation and explicit probability-sampling designs, a target law P is not established by the process through which scientific observations are collected. Units commonly enter through eligibility, recruitment, consent, availability, ascertainment, measurement, and curation rather than by independent selection from a prespecified target population.

¹Test-severity notions in the error-statistical tradition are defined as error probabilities under a sampling distribution in which the hypothesis is false; the thresholds here are not calibrated in that way and do not carry that meaning.

Even randomized experiments usually randomize treatment assignment among enrolled units, not the selection of those units from the population to which the result may later be generalized [29, 32]. Natural processes may of course contain stochastic elements, and a probability model may describe the collected observations post hoc, but neither fact identifies the law governing unrealized cases in the claimed scope. Consequently, $D \sim P^n$ is ordinarily a superpopulation conjecture connecting collected evidence to future observations, not a property established by the study design. Repartitioning D through cross-validation cannot test that conjecture or convert the realized collection process into sampling from P .

A second distinction concerns the conjectures that are tested. Cross-validation does not confront the final full-data fit

$$M^{(\text{all})} = A(D)$$

with withheld evidence. It constructs and evaluates

$$M^{(-1)} = A(D_{-1}), \dots, M^{(-K)} = A(D_{-K}),$$

each calibrated using less evidence than is available to the final fit. These fold-specific conjectures need not have worse realized performance in every case, but they are less fully informed products of the construction procedure. Their losses therefore bear directly on those reduced-data fits, and only indirectly on the model subsequently fitted to all of D . The resulting training-size and target mismatch is a recognized source of difficulty in cross-validated assessment [8, 4].

The mismatch becomes sharper when cross-validation participates in model construction. Fold outcomes may criticize candidate specifications and thereby guide the selection of variables, hyperparameters, or procedures. Once those outcomes have influenced the selected conjecture, the same cross-validation score cannot also serve as an untouched assessment of it. Nested cross-validation separates an inner construction process from an outer assessment and can reduce the resulting selection bias [30, 9]. It does not, however, cause the outer folds to test the final model fitted afterward on all available data. Cross-validation thus manages a genuine scarcity problem by rotating observations between construction and criticism; it does not abolish the distinction between them. Independent subsequent evidence permits the cleaner sequence: construct the strongest conjecture from the legitimate development evidence, then expose that completed conjecture to evidence unavailable during its construction.

When candidate models are intended as conjectures about a data-generating process, recurrence under a fixed P need not exhaust their claimed scope. Their comparative implications may also be intended to persist across prevalences, case mixtures, empirical contexts, or other scientifically named conditions. Expected predictive loss averages according to the stipulated law and therefore permits favorable parts of that law to compensate for failures elsewhere. The present framework addresses the broader comparison directly: it subjects completed model rankings to declared challenges and asks whether their oriented replacement claim survives. It does not model the data-generating process or establish either conjecture as true. Its advantage is that the form and interpretation of its evidence match a claim of comparative persistence, rather than replacing that claim with the expected performance of repeatedly refitted predictors under a presumed recurrence law.

7.4 Minimum reporting content

A concise report should identify the claim and the stage at which it survived or failed.

Table 2: Minimum reporting content for an anchored replacement analysis.

Component	Report
Claim and policy	Candidate and incumbent orientation; δ, d, γ ; reverse orientation; final verdict and whether it refers to the full or reduced assessment
Condition and effect	$\mathcal{C}$ and its basis; transport commitments; paired construction; tie convention; numerical condition search and tolerance
Empirical evidence	$\mathcal{R}$; admission and evaluation unit; contexts; known dependence or multiplicity; pooling and empirical selection rule; M and N_m ; observed anchors; K_E ; observed support and survival; number of anchors clearing d
Computational challenge	$\mathcal{B}$ and resampling unit; pairing and dependence preservation; J_m ; K_C ; whether row selections are independent; anchored support and survival; row survival fractions
Diagnostics	Binding anchors; computational effects below their anchors; values near d ; seed and J_m stability; sensitivity to prespecified alternative orders, conditions, and floors
Scope	Represented case types and contexts; absent mechanisms; whether a reference randomization is asserted; external premise required for deployment

A headline statement may read:

Across M admitted empirical evaluations satisfying the declared regime, direct order- K_E observed support was [value] and observed literal survival above $d = [\text{value}]$ was [value], so the comparison [passed/failed] the empirical gate. [If passed:] Each evaluation was then challenged using J_m paired computational replications under the declared procedure. With computational order K_C , observed-anchored nested support was [value] and nested literal survival was [value]. The full assessment therefore returned [supported replacement/no verdict] under $\delta = [\text{value}]$ and $\gamma = [\text{value}]$. The computational layer recombines represented units; it adds no empirical evaluations or unrepresented failure mechanisms.

If the gate fails, the report should state that anchor domination makes passage of the full rule impossible and that computation was unnecessary for the verdict. If Stage 2 was not declared, any passing verdict should be labeled support under the reduced observed-only assessment.

7.5 Beyond AP

The observed-anchored hierarchy applies to any metric that supplies condition-challenged observed and computational effects. Because AUROC is invariant to target prevalence, Appendix F instead challenges capped within-class redistributions of represented cases and applies the same nested replacement rule.

8 Conclusion

A favorable benchmark comparison does not by itself resolve whether a candidate should replace an incumbent. The measured advantage may depend on the target conditions under which performance is evaluated and on the empirical evaluations included in the comparison. A compensating aggregation, such as an average, can summarize these results while concealing that the candidate failed under a condition or evaluation the replacement claim was intended to cover. This paper therefore frames model replacement as a question about persistence: how much of the candidate’s paired advantage remains after it faces the conditions and empirical tests that the claim says it should survive?

The proposed strategy is to encode those commitments before the results are examined. Within each complete evaluation, the models are paired and the weakest advantage over the declared condition set is retained. The replacement rule then asks both how much advantage remains and how broadly it survives across the admitted empirical evidence. A staged computational extension can additionally probe dependence on the particular represented units without changing the empirical scope of the claim. For AP, the condition challenge concerns target prevalence; for AUROC, the same general strategy is applied to changes in within-class case mix.

This organization makes the evidential basis of a replacement verdict explicit and auditable. Passage can be traced to named conditions, empirical evaluations, challenge orders, thresholds, and computational procedures. Failure identifies where the available evidence ceased to support the oriented claim and returns no verdict rather than allowing the assessment to be redefined after the result. The paper therefore argues for an observed-anchored, noncompensating approach to model replacement grounded in the falsificationist tradition.

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A Empirical prior-standardized average precision

A.1 Definition and weighted implementation

Let

$$D = \{(s_i, y_i)\}_{i=1}^N, \quad y_i \in \{0, 1\},$$

with positive and negative counts $N_+ = \sum_i y_i > 0$ and $N_- = N - N_+ > 0$. Let $c_1 > \dots > c_H$ be the distinct score values. At threshold h , define

$$\text{TP}_h = \sum_i \mathbf{1}\{y_i = 1, s_i \geq c_h\}, \quad \text{FP}_h = \sum_i \mathbf{1}\{y_i = 0, s_i \geq c_h\},$$

and

$$r_h = \frac{\text{TP}_h}{N_+}, \quad f_h = \frac{\text{FP}_h}{N_-}.$$

With $r_0 = 0$, noninterpolated prior-standardized AP under the threshold-block convention is

$$\text{AP}_\pi(D) = \sum_{h=1}^H (r_h - r_{h-1}) \frac{\pi r_h}{\pi r_h + (1 - \pi) f_h}. \quad (28)$$

All units sharing a score enter one block. A negative-only block has no recall increment, but its false positives reduce precision in later blocks.

The same quantity follows by weighting each positive unit by π/N_+ and each negative unit by $(1 - \pi)/N_-$. Weighted precision at threshold h is the fraction in Equation (28), while weighted recall remains r_h .

A.2 Behavior across prevalence

Combining Equations (3) and (28) gives

$$\text{CNAP}_\pi(D) = \sum_{h=1}^H (r_h - r_{h-1}) \frac{\pi(r_h - f_h)}{\pi r_h + (1 - \pi) f_h}. \quad (29)$$

The derivative of one threshold contribution is

$$\frac{\partial}{\partial \pi} \left\{ \frac{\pi(r_h - f_h)}{\pi r_h + (1 - \pi) f_h} \right\} = \frac{f_h(r_h - f_h)}{\{\pi r_h + (1 - \pi) f_h\}^2}. \quad (30)$$

If $r_h \geq f_h$ at every threshold with positive recall increment, the marginal profile is nondecreasing. A paired difference may remain nonmonotone even when both marginal profiles are nondecreasing.

B Properties of finite-list and nested support

B.1 Finite-list survival and ordered representations

For $\mathbf{t} = (t_1, \dots, t_L)$ with entries in $[a, b]$, let

$$R_s = \sum_{\ell=1}^L \mathbf{1}\{t_\ell > s\}.$$

For each $I \in \mathcal{I}_{L,K}$,

$$\min_{\ell \in I} t_\ell = a + \int_a^b \mathbf{1}\{t_\ell > s \text{ for every } \ell \in I\} ds.$$

Exactly $\binom{R_s}{K}$ subsets have every effect above s . Averaging gives

$$S_K(\mathbf{t}) = a + \int_a^b \frac{\binom{R_s}{K}}{\binom{L}{K}} ds. \quad (31)$$

Thus finite-list literal survival at floor d is

$$\psi_{K,d}(\mathbf{t}) = \frac{\binom{R_d}{K}}{\binom{L}{K}}. \quad (32)$$

Let $t_{(1)} \leq \dots \leq t_{(L)}$. The minimum of exactly $\binom{L-i}{K-1}$ distinct order- K subsets is $t_{(i)}$. Hence

$$S_K(\mathbf{t}) = \binom{L}{K}^{-1} \sum_{i=1}^{L-K+1} \binom{L-i}{K-1} t_{(i)}. \quad (33)$$

This is an exact identity for the indexed finite list, not an estimate of a population spectral functional.

B.2 Proof of the nested magnitude–survival identity

Apply the layer-cake identity to each complete challenge value t :

$$t = a + \int_a^b \mathbf{1}\{t > s\} ds.$$

The challenge clears s exactly when, for every selected row, its observed anchor clears s and every selected computational effect clears s .

Conditional on row m , the fraction of computational selections having that property is $q_{m,K_C,s}^{\text{anc}}$. Under independent row selections, the fraction for empirical subset I is their product. Averaging over empirical subsets gives Equation (17). Averaging the layer-cake identity over all complete challenges and interchanging the finite average and integral proves Equation (19).

For coordinated row selections, replace the product by the survival probability under their declared joint selection law. The layer-cake identity continues to hold for that direct challenge definition.

B.3 Boundary reduction and proof of anchor domination

Under $K_C = 0$, the empty computational minimum is $+\infty$. Therefore the contribution from row m is $\min\{z_m^{\text{obs}}, +\infty\} = z_m^{\text{obs}}$, and the average over empirical subsets is precisely $S_{K_E}(\mathbf{z}^{\text{obs}})$. The corresponding complete challenge clears d precisely when all selected observed anchors clear it, giving Equation (20).

For every complete nested selection,

$$\min_{m \in I} \left\{ z_m^{\text{obs}}, \min_{j \in A_m} z_{mj}^{\text{comp}} \right\} \leq \min_{m \in I} z_m^{\text{obs}}.$$

Averaging proves Equation (22). If a complete nested selection clears d , its selected observed anchors necessarily clear d . The set of surviving nested selections is consequently contained in the set permitted by the observed anchors, proving Equation (23).

Removing the mandatory anchors replaces every selection minimum by a value no smaller than its anchored counterpart. The analogous domination of the unanchored nested projection follows for magnitude and survival.

B.4 Proof of hierarchical challenge monotonicity

Enlarging $\mathcal{C}$ lowers or preserves every observed and computational retained effect. Both anchored magnitude and literal survival are coordinatewise nondecreasing functions of those entries, so neither quantity can increase under the enlarged condition challenge.

For the computational order, every order- $K_C + 1$ subset contains $K_C + 1$ order- K_C subsets. Its anchored minimum is no larger than the average of their anchored minima. Averaging over all order- $K_C + 1$ subsets counts every order- K_C subset the same number of times. This proves within-row monotonicity. Applying the same argument simultaneously within the selected rows and then averaging across empirical subsets proves monotonicity of the full supported magnitude.

At a fixed floor, a surviving order- $K_C + 1$ computational subset has all of its order- K_C subsets surviving. Equivalently,

$$\frac{\binom{R}{K_C+1}}{\binom{J}{K_C+1}} \leq \frac{\binom{R}{K_C}}{\binom{J}{K_C}}$$

for $0 \leq R \leq J$. Thus every row survival fraction is nonincreasing in K_C .

The same subset-counting argument across empirical rows proves magnitude monotonicity in K_E . For survival, couple an order- $K_E + 1$ empirical subset with a uniformly selected order- K_E subset inside it. The product of row survival fractions in $[0, 1]$ cannot decrease when a factor is removed. Averaging the coupling proves monotonicity in K_E .

B.5 Proof of the nested orientation bound

For each full empirical or computational evaluation e , write

$$L_e = \inf_{c \in \mathcal{C}} \Delta_c^{A:B}(D_e), \quad U_e = \sup_{c \in \mathcal{C}} \Delta_c^{A:B}(D_e).$$

The retained effect in the reverse orientation is $-U_e$. For a fixed nested selection Q of observed anchors and computational entries, the forward and reverse challenge values are

$$\min_{e \in Q} L_e \quad \text{and} \quad -\max_{e \in Q} U_e.$$

Because $L_e \leq U_e$ for every entry,

$$\min_{e \in Q} L_e - \max_{e \in Q} U_e \leq 0.$$

Averaging over the same nested selections proves Equation (24).

B.6 Proof of the tolerated-count characterization

By Equation (32), observed literal survival is

$$\psi_{K_E, d}^{\text{obs}} = \frac{\binom{R}{K_E}}{\binom{M}{K_E}}.$$

Pascal's identity gives

$$\binom{R+1}{K_E} - \binom{R}{K_E} = \binom{R}{K_E - 1}.$$

Thus survival is nondecreasing in R and strictly increasing for $R \geq K_E - 1$. Because $\gamma > 0$, no value $R < K_E$ can satisfy the survival requirement. The passing values therefore form the upper set $\{r^*, \dots, M\}$, where

$$r^* = \min \left\{ r : \binom{r}{K_E} > \gamma \binom{M}{K_E} \right\}.$$

Hence the gate passes if and only if $R \geq r^*$, equivalently if at most $M - r^*$ admitted evaluations fail to clear the floor.

B.7 Noncomposition and preservation of index identity

For profiles U_c and V_c , define

$$U_{\mathcal{C}} = \inf_c U_c, \quad V_{\mathcal{C}} = \inf_c V_c, \quad T_{\mathcal{C}} = \inf_c (U_c - V_c).$$

Choosing a condition that approaches the infimum of U gives $T_{\mathcal{C}} \leq U_{\mathcal{C}} - V_{\mathcal{C}}$. Thus separately challenged marginal model effects dominate the correctly paired retained effect cellwise. Coordinatewise monotonicity propagates that discrepancy through the anchored hierarchy.

Row-summary composition fails because the minimum is concave. For independent row selections $A_1, \dots, A_M$,

$$\mathbb{E} \left[\min_{m \in I} \left\{ z_m^{\text{obs}}, \min_{j \in A_m} z_{mj}^{\text{comp}} \right\} \right] \leq \min_{m \in I} \mathbb{E} \left[\min \left\{ z_m^{\text{obs}}, \min_{j \in A_m} z_{mj}^{\text{comp}} \right\} \right].$$

The left side is the complete-challenge construction; the right side is what a row-first summary uses before the outer minimum. Strict inequality occurs in the running example.

Flattening can fail even more fundamentally. Suppose one empirical row has anchor 1 and all computational effects 1, while a second row has anchor -1 and all computational effects -1 . The observed order-1 support is zero, regardless of the Monte Carlo counts. A flattened mean becomes $(J_1 - J_2)/(J_1 + J_2)$ and may approach either endpoint merely by allocating more draws to one row. The flattened list therefore responds to approximation counts as though they were empirical evidence. Preserving the row index avoids that change of scientific meaning.

C Tie averaging and paired reference properties

C.1 Why ordinary pairing rewards arbitrary refinement

Take four units with two positive and two negative labels at $\pi = 1/2$. Model A assigns four distinct scores and model B assigns one common score. Assign the labels uniformly over the six possible

arrangements. Neither score vector contains outcome information. Under ordinary stepwise AP, model A has expected AP 49/72, while model B has AP 1/2 under every assignment. Therefore

$$\mathbb{E}[\text{CNAP}_A - \text{CNAP}_B] = \frac{13}{36}.$$

For two independently randomized label assignments, the expected order-2 minimum of the paired differences remains positive at 29/216. Pairing has favored the model that broke ties without information.

An additive reference correction cannot remove this displacement at every performance level. The randomization displacement is positive near no association but vanishes at perfect separation, where both models reach the ceiling. Refinement neutrality is therefore obtained only when outcome-blind tie handling occurs before condition and replication minima are applied.

C.2 Proof of refinement neutrality

Condition on S_B and Y . Under Proposition 1, model A 's strict ranking O_A is uniform on the rankings consistent with S_B . Tie-averaged AP for A is the conditional expectation of ordinary AP over whatever ties A retains. By the tower property,

$$\mathbb{E}[\widetilde{\text{AP}}_{A,\pi} \mid S_B, Y] = \mathbb{E}[\text{AP}_\pi(D_{O_A}) \mid S_B, Y].$$

Uniformity of O_A turns the right side into the average of ordinary AP over all strict rankings consistent with S_B , which is $\widetilde{\text{AP}}_{B,\pi}$.

C.3 Exchangeable-label reference behavior

The reference law below is a finite randomization device, not a superpopulation model. It places the uniform counting measure on the $\binom{N}{N_+}$ relabelings of the *observed* outcomes within each evaluation, conditional on the score vectors and class counts. This is the same kind of declared counting measure as the tie-resolution average of Section 2.4, taken over a finite, enumerable set of transformations of the actual data; no unobserved population is introduced [33, 35]. What distinguishes it from tie averaging is only that it permutes the outcomes, and so encodes a null of no score-label association. It is used descriptively—to record what the operator returns when labels carry no signal—and never as a significance statement about an observed value. None of the main results depends on it.

Definition 4 (Exchangeable-label reference law). *Conditionally on both score vectors and the positive count in each observed evaluation, every label vector with that count has equal probability. After a label vector is assigned, $\mathcal{B}$ is applied conditionally to that randomized evaluation, preserving pairing and the declared resampling structure.*

Fix N_+ , N_- , and a strict ranking. Under full exchangeability, the label sequence read along the ranking is uniform over all $\binom{N}{N_+}$ sequences with N_+ positives. Reading labels along any other strict ranking is a bijection of the same sequences. Expected tie-averaged CNAP therefore depends on the class counts and π , but not on the fixed score vector or its tie structure. Hence

$$\mathbb{E}_0[\Delta_\pi^{A:B}(D_m)] = 0 \tag{34}$$

for every condition-specific observed paired effect.

Because

$$\mathbb{E}_0 \left[\inf_{\pi \in \Pi_\star} \Delta_\pi^{A:B}(D_m) \right] \leq \inf_{\pi \in \Pi_\star} \mathbb{E}_0[\Delta_\pi^{A:B}(D_m)] = 0,$$

the expected observed retained anchor is nonpositive. Anchor domination gives

$$\Theta_{A:B;K_E,K_C}^{\text{anc}} \leq S_{K_E}(\mathbf{z}^{A:B,\text{obs}}) \leq \frac{1}{M} \sum_{m=1}^M z_m^{A:B,\text{obs}}.$$

Taking reference expectation proves that expected anchored support is nonpositive. Exchanging the model orientation proves the same result for $B : A$. This proof keeps the observed labels and the computational procedure in their proper order: labels are randomized within each empirical evaluation, then $\mathcal{B}$ is applied conditionally to that randomized evaluation.

Full exchangeability is stronger than independence. Take four units in two matched pairs, each constrained to contain one positive. Let model A tie all units, while model B places the first pair above the second and ties within pairs. At $\pi = 1/2$, exact tie averaging over the four design-permitted label assignments yields $\widehat{\text{CNAP}}_A - \widehat{\text{CNAP}}_B = 1/36$ in one orientation for every assignment. Restricted symmetry divides rankings into different orbits; tie averaging within blocks cannot move a ranking between them.

C.4 Why paired null centering fails

Take two positive and two negative units at $\pi = 1/2$, with

$$s_A = (4, 3, 2, 1), \quad s_B = (2, 2, 1, 1),$$

and let the two highest-scored units be positive. Both models separate the classes perfectly, so every stratified empirical-resampling evaluation has CNAP one for each model and paired difference zero.

Under joint label permutation, exact enumeration of unanchored order-2 support gives expected value $35/384$ for $A - B$ and $-295/1152$ for $B - A$. Subtracting these reference values would report a positive corrected value of about 0.256 for the second orientation, favoring the coarser model although both predictions are perfect. The displacement is not an additive constant; the reference law is therefore diagnostic rather than a centering correction.

D Closed-form computation of tie-averaged AP

Index distinct score blocks by $h = 1, \dots, H$ from highest to lowest. Let block h contain a_h positives and b_h negatives, with $m_h = a_h + b_h$, and let

$$A_h = \sum_{\ell < h} a_\ell, \quad B_h = \sum_{\ell < h} b_\ell.$$

Then

$$\begin{aligned} \widetilde{\text{AP}}_\pi(D) &= \sum_{h=1}^H \frac{a_h}{N_+} \frac{1}{m_h} \sum_{k=1}^{m_h} \sum_{t=\max(0, k-1-b_h)}^{\min(k-1, a_h-1)} \frac{\binom{a_h-1}{t} \binom{b_h}{k-1-t}}{\binom{m_h-1}{k-1}} \\ &\quad \times \frac{\pi r_{hkt}}{\pi r_{hkt} + (1-\pi) f_{hkt}}, \end{aligned} \tag{35}$$

where

$$r_{hkt} = \frac{A_h + t + 1}{N_+}, \quad f_{hkt} = \frac{B_h + k - 1 - t}{N_-}.$$

To obtain the formula, fix one positive unit in block h . Under a uniformly random ordering of the block, its position k is uniform on $\{1, \dots, m_h\}$. Given k , the number t of other positive units in the $k - 1$ preceding positions is hypergeometric with $a_h - 1$ positives and b_h negatives available. The last fraction is prior-standardized precision when that positive enters the ranking. Averaging over t and k , multiplying by the a_h positives in the block, and summing across blocks gives the result.

The calculation costs $O(\sum_h m_h^2)$ operations. Its combinatorial weights do not depend on π and can be reused throughout the prevalence search. When all scores are distinct, Equation (35) reduces to ordinary stepwise AP.

E Computation for several empirical evaluations

E.1 Observed gate

For each admitted empirical evaluation, compute

$$z_m^{\text{obs}} = \inf_{c \in \mathcal{C}} \Delta_c(D_m).$$

Sort the anchors as $z_{(1)}^{\text{obs}} \leq \dots \leq z_{(M)}^{\text{obs}}$. Then

$$S_{K_E}(\mathbf{z}^{\text{obs}}) = \binom{M}{K_E}^{-1} \sum_{i=1}^{M-K_E+1} \binom{M-i}{K_E-1} z_{(i)}^{\text{obs}}. \quad (36)$$

Observed literal survival follows from the count of anchors above d . If the gate fails, the primary algorithm stops.

E.2 Finite-array sweep

If the gate passes, generate each row's computational effects. Direct enumeration requires considering

$$\sum_{I \in \mathcal{I}_{M, K_E}} \prod_{m \in I} \binom{J_m}{K_C}$$

complete selections and quickly becomes infeasible. The survival representation avoids this enumeration.

At each effect level s , compute the row fractions in Equation (16), then evaluate the elementary symmetric polynomial in Equation (18). Its recursion is

$$e_0^{(m)} = 1, \quad e_k^{(m)} = e_k^{(m-1)} + q_m(s) e_{k-1}^{(m-1)}, \quad (37)$$

for $k = 1, \dots, K_E$, updated downward in k for each row. One survival evaluation costs $O(MK_E)$.

The survival curve changes only when s crosses an observed or computational effect. Sorting the distinct effect levels and summing the area of the piecewise-constant survival curve gives the exact finite-array supported magnitude in Equation (19). The complete calculation may proceed through the following sweep:

1. collect and sort every distinct observed and computational effect level;
2. initialize $R_{m,s}$ above the largest level and update row counts as the sweep crosses computational effects;
3. set $q_m(s) = 0$ whenever the sweep has reached or exceeded the observed anchor, and otherwise calculate the binomial ratio in Equation (16);
4. use the recursion in Equation (37) to obtain e_{K_E} and hence nested survival; and
5. multiply each constant survival value by the width to the next effect level and sum the areas, adding the lower endpoint a .

Using log-binomial evaluations or stable multiplicative ratios avoids overflow when J_m is large. Because only exceedance counts enter, repeated equal effects should be updated as one threshold block. The strict choice $>$ rather than $\geq$ should match the declared literal-survival convention.

For the one-evaluation boundary, a simpler ordered representation is available. Sort $t_1, \dots, t_J$ as $t_{(1)} \leq \dots \leq t_{(J)}$. Then

$$S_{1;K_C}^{\text{anc}}(u; \mathbf{t}) = \binom{J}{K_C}^{-1} \sum_{i=1}^{J-K_C+1} \binom{J-i}{K_C-1} \min\{u, t_{(i)}\}. \quad (38)$$

E.3 A deliberately favorable ProteinGym stress test

The declared ProteinGym analyses stop at Stage 1, leaving the full $M > 1, K_E \geq 2, K_C > 0$ calculation without a real-data example. To illustrate that calculation, we searched over restricted target-prevalence intervals for one under which ESM-1v ensemble minus ESM-1v single would clear the observed gate. The interval $[0.40, 0.50]$ did so with $K_E = K_C = 2$, $\delta = d = 0$, and the illustrative breadth requirement $\gamma = 0.60$. This specification was constructed to expose Stage 2 rather than asserted as a prospective ProteinGym replacement policy.

Across the 91 assays, the observed anchors had mean 0.022877 and order-2 supported magnitude 0.000590. Seventy-four anchors exceeded zero, so

$$\frac{\binom{74}{2}}{\binom{91}{2}} = \frac{2,701}{4,095} = 0.659585$$

of the distinct assay pairs survived. The deliberately favorable specification therefore passed Stage 1.

We then generated $J = 200$ paired computational evaluations within each assay by sampling its observed number of residue positions with replacement and carrying every retained substitution at each selected position. A complete draw lacking either outcome class was redrawn. Each computational evaluation yielded one retained effect after the complete prevalence challenge. A complete nested challenge selected two assays, retained their two observed anchors, selected two computational effects within each assay, and took the minimum over all six effects. The finite array represents

$$\binom{91}{2} \binom{200}{2}^2 = 1,621,660,950,000$$

complete selections, whose aggregate is obtained by the sweep above without enumerating them individually.

Observed-anchored supported magnitude fell to -0.016395 , and literal survival fell to 0.457419 . Both requirements failed, so the full calculation returned no verdict at Stage 2. Thus, even after narrowing the prevalence range specifically to pass Stage 1—and using a less demanding breadth requirement than the primary ProteinGym policy—the advantage did not survive the within-assay computational challenge. Even in this deliberately favorable construction, the finite evidence did not support a robust superiority claim for one ESM-1v specification over the other.

E.4 Numerical and Monte Carlo checks

For a finite condition set, the condition challenge is exact. A grid used only to approximate a continuous condition set can miss an interior minimum and raise the retained effect. For each cell, a one-dimensional condition optimizer should therefore be checked at interval endpoints, against a dense grid, and at any analytically available stationary points. Reports should record the optimizer tolerance and whether the displayed limiting condition is unique.

Nonunique or moving minimizers require a distinction between the retained effect value and the identity of its limiting condition. Uniformly accurate profiles can yield an accurate infimum even when its location is unstable. Reports should therefore separate numerical accuracy of the values, stability of their minimizers, Monte Carlo stability across arrays, and policy sensitivity when effects lie near d .

For the computational array, repeat the calculation under independent seeds and increasing row counts J_m . Compare the anchored magnitude, nested survival at d , binding anchors, row survival fractions, and verdict. A stable unanchored projection is insufficient if anchoring changes which effect levels control the full assessment. Because a one-list order-2 standard-error formula does not automatically transfer to the full hierarchy, a hierarchical influence or U-statistic standard error should be used only after its applicability to the declared joint resampling law has been established.

When J_m differs between rows, each row’s distinct-subset fraction estimates the same declared rowwise target only if the resampling procedures and K_C are those specified by the assessment. The construction keeps the outer empirical-row weighting equal when approximation counts differ.

F Observed-anchored empirical case-mix breakdown for AUROC

F.1 The condition challenge for AUROC

AUROC is the probability that a randomly drawn positive outranks a randomly drawn negative, with half credit for a tie. It depends on the two class-conditional score distributions and is unchanged by pure prior shift. Target prevalence therefore supplies no AUROC challenge.

Within-class composition may still change. When named target mixtures are not available, the observed cases can support a deterministic empirical concentration challenge. The construction below admits increasingly concentrated redistributions over represented positive and negative cases and records where the replacement policy first fails. It does not claim that any admissible redistribution is an unseen population.

Let empirical evaluation D contain paired scores (s_i^A, s_i^B) and outcome Y_i , with positive index set $\mathcal{I}_+$ and negative index set $\mathcal{I}_-$. For $i \in \mathcal{I}_+$ and $k \in \mathcal{I}_-$, define

$$\phi_m(i, k) = \mathbf{1}\{s_i^m > s_k^m\} + \frac{1}{2}\mathbf{1}\{s_i^m = s_k^m\}, \quad m \in \{A, B\},$$

and the paired gain matrix

$$G_{ik} = \phi_A(i, k) - \phi_B(i, k).$$

For normalized within-class weights w_+ and w_- , the paired AUROC difference is

$$\Delta_D(w_+, w_-) = \sum_{i \in \mathcal{I}_+} \sum_{k \in \mathcal{I}_-} w_{+,i} w_{-,k} G_{ik}. \quad (39)$$

Uniform weights recover the ordinary empirical paired AUROC difference. Both models face the same weights, so the comparison remains paired. No AP-style tie correction is needed because every positive–negative pair already gives half credit for a tie.

Unrestricted weights degenerate to the least favorable single matrix entry. To control concentration, let N_+ and N_- be the class counts and define, for common factor $\Gamma \geq 1$,

$$\mathcal{W}_{+,\Gamma}(D) = \left\{ w_+ : \sum_i w_{+,i} = 1, \ 0 \leq w_{+,i} \leq \frac{\Gamma}{N_+} \right\}, \quad (40)$$

$$\mathcal{W}_{-,\Gamma}(D) = \left\{ w_- : \sum_k w_{-,k} = 1, \ 0 \leq w_{-,k} \leq \frac{\Gamma}{N_-} \right\}. \quad (41)$$

At $\Gamma = 1$, both weight vectors are uniform. Larger values permit greater concentration. For either class y ,

$$\text{ESS}(w_y) = \left(\sum_i w_{y,i}^2 \right)^{-1} \geq \frac{N_y}{\Gamma}.$$

Thus $1/\Gamma$ is a guaranteed minimum effective-support fraction implied by the pointwise cap.

The retained AUROC effect in full evaluation D is

$$Z_\Gamma^{A:B}(D) = \inf_{\substack{w_+ \in \mathcal{W}_{+,\Gamma}(D) \\ w_- \in \mathcal{W}_{-,\Gamma}(D)}} \Delta_D(w_+, w_-). \quad (42)$$

This is the condition-level infimum. The sets are nested in Γ , so Z_Γ is nonincreasing. At $\Gamma = 1$, it equals the ordinary paired empirical AUROC difference.

F.2 Anchors, computational effects, and the breakdown profile

For empirical evaluation D_m , define its observed AUROC anchor at concentration factor Γ as

$$z_m^{\text{obs}}(\Gamma) = Z_\Gamma^{A:B}(D_m).$$

For computational evaluation $D_m^{(j)}$ generated under $\mathcal{B}$, define

$$z_{mj}^{\text{comp}}(\Gamma) = Z_\Gamma^{A:B}(D_m^{(j)}).$$

The case-mix optimization is performed inside every full empirical or computational evaluation. The resulting anchors and cells then enter the same hierarchy as the AP effects.

At each common Γ , set

$$\Theta_{A:B;K_E,K_C}^{\text{anc}}(\Gamma) = S_{K_E;K_C}^{\text{anc}}\left(\mathbf{z}^{A:B,\text{obs}}(\Gamma); \mathbf{Z}^{A:B,\text{comp}}(\Gamma)\right), \quad (43)$$

$$\Psi_{A:B;K_E,K_C,d}^{\text{anc}}(\Gamma) = \Psi_{K_E;K_C,d}^{\text{anc}}\left(\mathbf{z}^{A:B,\text{obs}}(\Gamma); \mathbf{Z}^{A:B,\text{comp}}(\Gamma)\right). \quad (44)$$

The same concentration allowance applies to every evaluation, while the adverse weights may differ between evaluations and computational cells. Pointwise monotonicity of every effect makes both nested profiles nonincreasing in Γ .

At $\Gamma = 1$, first apply the observed empirical gate. If it fails, the anchored policy has no supported replacement before case-mix stress is added. If it passes and a computational procedure is declared, apply the full anchored rule at $\Gamma = 1$ before searching for breakdown.

Definition 5 (Anchored empirical case-mix breakdown factor). *Suppose the full anchored AUROC rule passes at $\Gamma = 1$. Define*

$$\Gamma_{\text{ECM},A:B}^{\dagger,\text{anc}} = \inf \left\{ \Gamma \geq 1 : \Theta_{A:B;K_E,K_C}^{\text{anc}}(\Gamma) \leq \delta \text{ or } \Psi_{A:B;K_E,K_C,d}^{\text{anc}}(\Gamma) \leq \gamma \right\}. \quad (45)$$

The factor is the first common upper within-class concentration allowance at which either replacement requirement fails. If the observed gate fails at $\Gamma = 1$, report no supported replacement at the observed mix rather than a computational breakdown search. If the observed gate passes but the full anchored rule fails at one, the breakdown under the declared full assessment is one. If no failure occurs before the finite weight sets saturate, report that no breakdown was found over the redistributions permitted by the represented cases.

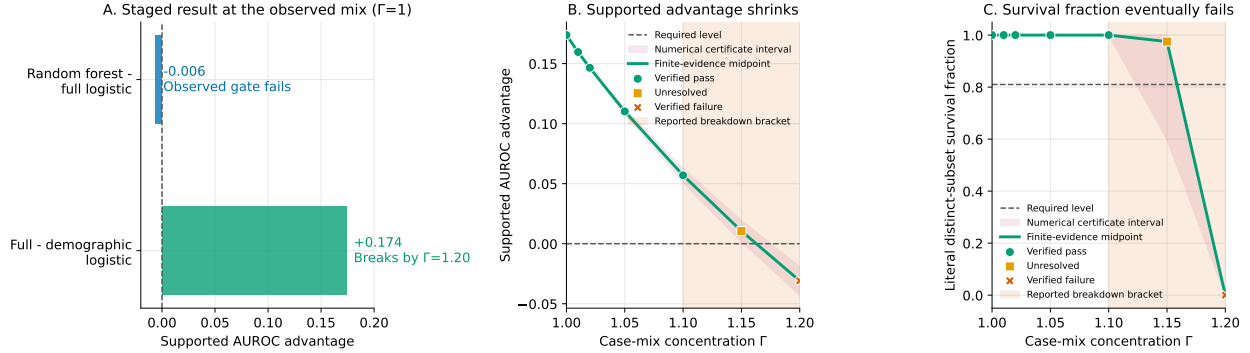
When no computational layer is declared, set $K_C = 0$ and obtain the reduced observed-only breakdown. When $M = 1$, $K_E = 1$ and the observed AUROC effect remains a mandatory anchor throughout the computational and concentration challenges.

F.3 Empirical AUROC companions

The Bank Marketing and ProteinGym data used in Section 6 also illustrate the two boundary cases of the AUROC construction. These calculations reuse the paired scores, evaluation units, orientations, and replacement thresholds already declared for those illustrations. Numerical lower and upper bounds below are optimization certificates, not statistical confidence intervals.

One empirical evaluation with a computational challenge. For Bank Marketing, random forest minus full logistic regression had ordinary paired AUROC difference -0.0059 and failed the observed gate at $\Gamma = 1$. The positive-control comparison of full logistic regression minus demographic logistic regression had ordinary paired AUROC difference 0.1787 . Its full observed-anchored order-2 assessment at $\Gamma = 1$ retained supported magnitude 0.1738 and literal survival 1.000 across $J = 80$ same-design computational replications. It therefore entered the case-mix search. The last verified pass occurred at $\Gamma = 1.10$, $\Gamma = 1.15$ remained numerically unresolved, and the first verified failure occurred at $\Gamma = 1.20$, where both policy requirements failed. The reported result is consequently the bracket $1.10 < \Gamma_{\text{ECM}}^{\dagger,\text{anc}} \leq 1.20$, not a point estimate of the breakdown factor.

How much case-mix change overturns the AUROC decision?



$\Gamma=1$ is the observed case mix; larger Γ permits greater concentration on fewer cases.

Figure 7: Observed-anchored AUROC case-mix breakdown in the Bank Marketing temporal evaluation. Panel A applies the observed gate before the computational challenge. Panels B and C show the supported-magnitude and literal-survival profiles for the positive control under increasing within-class concentration. The ribbon is the numerical certificate interval; markers distinguish verified passes, unresolved values, and verified failures. The shaded region is the reported breakdown bracket. This is the $M = 1$, $K_E = 1$, $K_C = 2$, $J = 80$ projected boundary of the full construction.

Several empirical evaluations without a computational challenge. For ProteinGym, each assay contributed its ordinary paired AUROC difference at $\Gamma = 1$. Applying the declared order-2 empirical challenge gave support -0.0663 for ESM-1v ensemble minus EVE ensemble, -0.0445 for ESM-2 650M minus ESM-1v ensemble, and -0.0659 for ESM-2 650M minus EVE ensemble. The numbers of positive assay effects were respectively 56, 48, and 56 of 91, giving exact assay-pair survival fractions 0.376, 0.275, and 0.376. All three comparisons therefore failed both direct requirements at the observed mix. The staged rule stopped without within-assay resampling or an AUROC concentration search. This is the observed-only $M = 91$, $K_E = 2$, $K_C = 0$ boundary and does not fill the full nested interior.

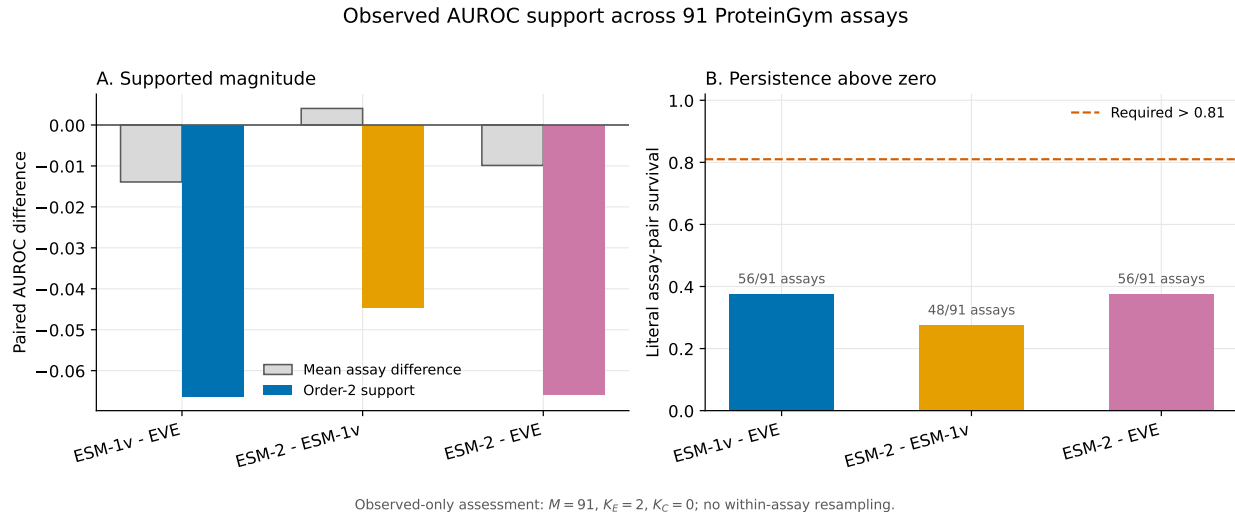


Figure 8: Observed-only AUROC support across the 91 ProteinGym assays. Panel A compares the mean paired assay difference with order-2 supported magnitude. Panel B gives the exact fraction of the $\binom{91}{2} = 4,095$ distinct assay pairs whose two AUROC effects both strictly exceed zero. All comparisons fail the declared survival requirement. No within-assay computational resampling is performed.

F.4 Computation and reporting

Equation (42) minimizes the bilinear form $w_+^\top G w_-$ over two capped simplices. A feasible pair of weights supplies an upper bound on the infimum and can demonstrate policy failure. Establishing that the policy continues to hold requires a valid global lower bound; alternating optimization without such a bound may stop above the true infimum and overstate survival.

Monotonicity permits a bracketed search over Γ . Within every searched value, the global optimization is repeated for each observed anchor and each required computational cell. The same computational evaluation $D_m^{(j)}$ is reused throughout the search so that changes in the profile arise from the concentration allowance rather than from different resamples.

A minimum report gives the Stage 1 result at $\Gamma = 1$, the Stage 2 result when applicable, the breakdown factor, the first failing requirement, the search tolerance, the global-optimization gap, and Monte Carlo stability for a computational array. The factor should be labeled full anchored or reduced observed-only.

Every admissible weight vector redistributes cases already present in an evaluation. A type absent from that evaluation has weight zero at every Γ . Individual reweighting may also concentrate on sample-specific errors rather than a named subpopulation. The breakdown factor consequently measures survival under a reproducible empirical concentration rule; it is not the magnitude of a known deployment shift.

F.5 Certified global optimization

Because the pass, unresolved, and failure labels in Appendix F.3 depend on the validity of a global lower bound, we record how the certificates were computed. The retained effect minimizes the bilinear form $w_+^\top G w_-$ over two capped simplices. Writing the cap as excluded mass— E on the positive cases and H on the negative cases, with total excluded mass $n_+(1-1/\Gamma)$ and $n_-(1-1/\Gamma)$ —the objective separates into three terms linear in the excluded sets and a single cross term that couples the two classes. The linear terms are minimized exactly by a greedy fill, because a linear objective over a capped simplex attains its minimum at a vertex that excludes the most adverse cases first. The cross term is bounded in absolute value by the product of the excluded masses, since each paired gain entry lies in $\{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}$.

A feasible weighting is obtained by alternating best response: minimize over the positive weights with the negative weights fixed, then reverse. Every iterate is feasible, so this returns a valid upper bound and can demonstrate that a requirement fails; it cannot certify that a requirement holds, which requires a valid lower bound. For the lower bound we take the larger of two complementary relaxations. The first drops the cross term, at a cost equal to $(\Gamma - 1)^2$ in the normalized scale, and is tight near $\Gamma = 1$. The second lets each retained case face its own worst opposing cases, relaxing the requirement of a common opposing weighting; it is weak near $\Gamma = 1$, strengthens as Γ grows, and becomes exact once a cap saturates. Because the gain matrix takes only five values, each case’s relaxed value is read from a five-bin histogram in constant time. When these bounds do not already settle the decision, a verification-only branch and bound narrows the interval under a node budget (here 50,000); exhausting the budget widens the reported interval rather than declaring a value optimal.

Certification is one-sided and decision-directed. The search halts as soon as the certified interval lies strictly on one side of the tested floor: a verified pass when the lower bound clears it, a verified failure when the upper bound falls below it, and *unresolved* when the interval straddles it within budget. The reported gaps are the absolute gap, upper minus lower, and the relative gap, the absolute gap divided by $\max(|\text{upper}|, 1)$; the certificates use an absolute and relative gap tolerance of 10^{-9} . At $\Gamma = 1$, and once both caps saturate, the optimum is closed-form and no search is required. The Bank Marketing bracket $1.10 < \Gamma_{\text{ECM}}^{\dagger, \text{anc}} \leq 1.20$ is exactly a verified pass at $\Gamma = 1.10$, an unresolved interval at $\Gamma = 1.15$, and a verified failure at $\Gamma = 1.20$; the breakdown search brackets the factor on a geometric ladder in Γ and then bisection, and an interval that cannot be tightened below the search tolerance is reported as a bracket rather than a point.

Two independent backends compute the same quantity: the combinatorial solver just described, and a mixed-integer linearization, solved with HiGHS [34], that introduces one product variable per case pair and serves as a reference on small problems. Both are checked against exhaustive enumeration of the capped-simplex vertex pairs on small evaluations.²

²The solver and the differential tests are provided in the accompanying source (the `auroc` module).